Technical data sheet

MULTIline PRO 4.5 mm

Hose liner for the trenchless inner lining of defective, leaking and statically impaired pipes.

Product features	
Product name	Lateral Repairs MULTIline PRO 4.5 mm
Product code	PRO
Length	50 m; 100 m / 164 feet; 328 feet
Diameter	70, 80, 100, 125, 150, 200, 225, 250, 300.
Wall thickness	4,50* mm / 0,18* inch
Liner undersized	13 %
Heat resistance	Max. 70 °C / 158 °F
Negotiating bends	Max. 90°

Material features	
Material	Multi (knitted) fleece with one sided TPU.
Textile	Polyester
Textile weight	550* g/m^2
Coating	One side TPU
Coating weight	150* g/m^2
Colour / coating	White / Transparent
Water penetration	≤ 550 mbar / ≤ 7,97 psi
Storage	Protected from light, dry

Features			Length		Resin	Pressure		Roller
Dimension	Flat (mm)	Bend	L + *	L + **	kg / m	Inversion (bar)	3D (bar)	Roller gap
DN 70-100	~95	90°	5 cm	5 %	~0.82	0,50-0,60	0,60-0,75	9mm
DN 100-150	~136	90°	5 cm	5 %	~1.15	0,40-0,55	0,45-0,60	9mm
DN 150-200	~205	90°	5 cm	5 %	~1.70	0,35-0,45	0,40-0,50	9mm
DN 200-250	~273	90°	5 cm	5 %	~2.45	0,30-0,35	0,30-0,40	9mm
DN 250-300	~341	90°	5 cm	5 %	~2.99	0,25-0,35	0,30-0,40	9mm

Additional length *per bend >45° **Approx 5 % loss in length for every 25 % change in dimension

Resin quantity

Calculate the exact resin amount with the free Lateral Repairs app.

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Notice

- Given the diversity of installation conditions, application areas and process techniques, the information in this data sheet can only be regarded as non-binding guideline values.
- Length addition: add the stated percentage of the liner length, plus the cm value per 45° bend; calculate the exact resin quantity with the Lateral Repairs app (scan the QR codes).
- The final quality depends on the resin system used, the inversion pressure and the curing pressure. We advise against the use of other resin systems or higher pressures and accept no liability.
- All data were determined at 20 °C / 68 °F and are bench-scale values that can differ on industrial job sites.

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Technical data sheet

MULTIline PRO 5.5 mm

Hose liner for the trenchless inner lining of defective, leaking and statically impaired pipes.

Product features	
Product name	Lateral Repairs MULTIline PRO 5.5 mm
Product code	PRO
Length	50 m; 100 m / 164 feet; 328 feet
Diameter	70, 80, 100, 125, 150, 200, 225, 250, 300.
Wall thickness	5,50 mm / 0,22 inch
Liner undersized	13 %
Heat resistance	Max. 70 °C / 158 °F
Negotiating bends	Max. 90°

Material features	
Material	Multi knitted Fleece with One - Sided TPU Coating.
Textile	Polyester
Textile weight	800* g/m²
Coating	One side TPU
Coating weight	150* g/m²
Colour / coating	White / Transparent
Water penetration	≤ 600 mbar / ≤ 8,70 psi
Storage	Protected from light, dry

Features			Length		Resin	Pressure		Roller
Dimension	Flat (mm)	Bend	L + *	L + **	kg / m	Inversion (bar)	3D (bar)	Roller gap
DN 70-100	~95	90°	5 cm	5 %	~0.95	0,55-0,65	0,60-0,75	12mm
DN 100-150	~136	90°	5 cm	5 %	~1.32	0,40-0,60	0,40-0,65	12mm
DN 150-200	~205	90°	5 cm	5 %	~1.88	0,40-0,50	0,40-0,55	12mm
DN 200-250	~273	90°	5 cm	5 %	~2.65	0,30-0,40	0,30-0,45	12mm
DN 250-300	~341	90°	5 cm	5 %	~3.25	0,25-0,35	0,35-0,45	12mm

Additional length *per bend >45° **Approx 5 % loss in length for every 25 % change in dimension

Resin quantity

Calculate the exact resin amount with the free Lateral Repairs app.

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Notice

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- Length addition: add the stated percentage of the liner length, plus the cm value per 45° bend; calculate the exact resin quantity with the Lateral Repairs app (scan the QR codes).
- The final quality depends on the resin system used, the inversion pressure and the curing pressure. We advise against the use of other resin systems or higher pressures and accept no liability.
- All data were determined at 20 °C / 68 °F and are bench-scale values that can differ on industrial job sites.

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Technical data sheet

MULTIline FLEX

Flexible hose liner for the trenchless rehabilitation of drains and house connections.

Product features	
Product name	Lateral Repairs MULTIline FLEX 3.0mm
Product code	FLEX
Length	50 m; 100 m / 164 feet; 328 feet
Diameter	30, 40, 50, 60, 65, 70, 75, 80, 100, 125, 150, 200, 225, 250.
Wall thickness	3,50* mm / 0,14* inch
Liner undersized	5 %, 9 %, 18 %
Heat resistance	Max. 50 °C / 122 °F
Negotiating bends	90°

Material features	
Material	PES-Plush (knitted) and brushed with one sided PU-Coating.
Textile	Polyester
Textile weight	450* g/m²
Coating	One side PU
Coating weight	250* g/m²
Colour / coating	White / Transparent
Water penetration	≤ 500 Mbar / ≤ 7,25 psi
Storage	Protected from light, dry

Features			Length		Resin	Pressure		Roller
Dimension	Flat (mm)	Bend	L + *	L + **	kg / m	Inversion (bar)	3D (bar)	Roller gap
DN 30-50	~42	90°	5 cm	5 %	~0.32	0,55-0,70	0,75-0,95	7mm
DN 50-70	~71	90°	5 cm	5 %	~0.51	0,50-0,65	0,60-0,95	7mm
DN 70-100	~100	90°	5 cm	5 %	~0.73	0,50-0,60	0,55-0,90	7mm
DN 100-150	~143	90°	5 cm	5 %	~1.01	0,45-0,55	0,50-0,85	7mm
DN 125-175	~178	90°	5 cm	5 %	~1.24	0,40-0,50	0,45-0,70	7mm
DN 150-200	~214	90°	5 cm	5 %	~1.58	0,35-0,45	0,40-0,65	7mm
DN 200-250	~285	90°	5 cm	5 %	~2.08	0,30-0,40	0,35-0,55	7mm
DN 225-275	~320	90°	5 cm	5 %	~2.31	0,30-0,35	0,35-0,45	7mm
DN 250-300	~357	90°	5 cm	5 %	~2.58	0,25-0,30	0,35-0,40	7mm

Additional length *per bend >45° **Approx 5 % loss in length for every 25 % change in dimension

Resin quantity

Calculate the exact resin amount with the free Lateral Repairs app.

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Notice

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- Length addition: add the stated percentage of the liner length, plus the cm value per 45° bend; calculate the exact resin quantity with the Lateral Repairs app (scan the QR codes).
- The final quality depends on the resin system used, the inversion pressure and the curing pressure. We advise against the use of other resin systems or higher pressures and accept no liability.
- All data were determined at 20 °C / 68 °F and are bench-scale values that can differ on industrial job sites.

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Technical data sheet

MULTIline CORE

Multi-knitted hose liner for the trenchless inner lining of pipes.

Product features	
Product name	Lateral Repairs MULTILine CORE
Product code	CORE
Length	50 m; 100 m / 164 feet; 328 feet
Diameter	70, 80, 100, 125, 150, 200, 225, 250.
Wall thickness	4,50* mm / 0,18* inch
Liner undersized	13 %
Heat resistance	Max. 85 °C / 167 °F
Negotiating bends	Max. 90°

Material features	
Material	Multi-knitted fleece with one sided Polypropylene (PP) coating.
Textile	Polyester
Textile weight	550* g/m²
Coating	One side PP
Coating weight	300* g/m²
Colour / coating	White / Transparent
Water penetration	≤ 550 mbar / ≤ 7,97 psi
Storage	Protected from light, dry

Features			Length		Resin	Pressure		Roller
Dimension	Flat (mm)	Bend	L + *	L + **	kg / m	Inversion (bar)	3D (bar)	Roller gap
DN 70-100	~95	90°	5 cm	4 %	~0.94	0,50-0,60	0,60-0,75	8mm
DN 100-150	~136	90°	5 cm	4 %	~1.25	0,40-0,55	0,45-0,60	8mm
DN 150-200	~205	90°	5 cm	4 %	~1.90	0,35-0,45	0,40-0,50	8mm
DN 200-250	~273	90°	5 cm	4 %	~2.60	0,30-0,35	0,30-0,40	8mm
DN 250-300	~341	90°	5 cm	4 %	~3.10	0,25-0,35	0,30-0,40	8mm
DN 300-350	~408	90°	5 cm	4 %	~3.80	0,20-0,30	0,25-0,35	8mm

Additional length *per bend >90° **Approx 4 % loss in length for every 25 % change in dimension

Resin quantity

Calculate the exact resin amount with the free Lateral Repairs app.

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Notice

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Technical data sheet

MULTIline FORCE

High-strength, reinforced hose liner for the structural rehabilitation of pipes.

Product features	
Product name	Lateral Repairs MULTIline FORCE 3.0mm
Product code	FORCE
Length	50 m; 100 m / 164 feet; 328 feet
Diameter	100 mm – 300 mm / 4 inch – 12 inch
Wall thickness	3,00 mm / 0,12 inch
Liner undersized	10 %
Heat resistance	Max. 100 °C / 212 °F
Negotiating bends	Max. 45°

Material features	
Material	Polyester needle-punched fleece liner with filament reinforcement and a polypropylene (PP) coating.
Textile	Polyester
Textile weight	650 g/m²
Coating	PP
Coating weight	300 g/m²
Colour / coating	White / Milky white
Water penetration	≥ 500 mbar
Storage	Protected from light, dry

Features			Length		Resin	Pressure		Roller
Dimension	Flat (mm)	Bend	L + *	L + **	kg / m	Inversion (bar)	3D (bar)	Roller gap
DN 100	~140	2 x 45	~3 cm	–	~ 1.15	0,35-0,40	–	8mm
DN 125	~175	2 x 45	~3 cm	–	~1.44	0,30-0,45	–	8mm
DN 150	~212	2 x 45	~3 cm	–	~1.72	0,30-0,50	–	8mm
DN 200	~282	2 x 45	~3 cm	–	~2.30	0,25-0,40	–	8mm
DN 225	~318	2 x 45	~3 cm	–	~2.60	0,20-0,40	–	8mm
DN 250	~353	2 x 45	~3 cm	–	~2.90	0,20-0,40	–	8mm
DN 300	~424	2 x 45	~3 cm	–	~3.21	0,20-0,40	–	8mm

Additional length *per bend >45°

Resin quantity

Calculate the exact resin amount with the free Lateral Repairs app.

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Notice

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- Length addition: add the stated percentage of the liner length, plus the cm value per 45° bend; calculate the exact resin quantity with the Lateral Repairs app (scan the QR codes).
- The final quality depends on the resin system used, the inversion pressure and the curing pressure. We advise against the use of other resin systems or higher pressures and accept no liability.
- All data were determined at 20 °C / 68 °F and are bench-scale values that can differ on industrial job sites.

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Technical data sheet

MULTIline FORCE RF

High-strength, reinforced hose liner for the structural rehabilitation of pipes.

Product features	
Product name	Lateral Repairs MULTIline FORCE 4.5mm
Product code	FORCE
Length	50 m; 100 m / 164 feet; 328 feet
Diameter	100 mm – 600 mm / 4 inch – 24 inch
Wall thickness	4,50 mm / 0,18 inch
Liner undersized	10 %
Heat resistance	Max. 100 °C / 212 °F
Negotiating bends	Max. 45°

Material features	
Material	Polyester needle-punched fleece liner with filament reinforcement and a polypropylene (PP) coating.
Textile	Polyester
Textile weight	900 g/m²
Coating	PP
Coating weight	300 g/m²
Colour / coating	White / Milky white
Water penetration	≥ 500 mbar
Storage	Protected from light, dry

Features			Length		Resin	Pressure		Roller
Dimension	Flat (mm)	Bend	L + *	L + **	kg / m	Inversion (bar)	3D (bar)	Roller gap
DN 100	~140	2 x 45	~3 cm	–	~1.60	0,35-0,40	–	11mm
DN 125	~175	2 x 45	~3 cm	–	~2.00	0,30-0,45	–	11mm
DN 150	~212	2 x 45	~3 cm	–	~2.41	0,30-0,50	–	11mm
DN 200	~282	2 x 45	~3 cm	–	~3.21	0,25-0,40	–	11mm
DN 225	~318	2 x 45	~3 cm	–	~3.61	0,20-0,40	–	11mm
DN 250	~353	2 x 45	~3 cm	–	~4.01	0,20-0,40	–	11mm
DN 300	~424	2 x 45	~3 cm	–	~4.81	0,20-0,40	–	11mm

Additional length *per bend >45°

Resin quantity

Calculate the exact resin amount with the free Lateral Repairs app.

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Notice

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- Length addition: add the stated percentage of the liner length, plus the cm value per 45° bend; calculate the exact resin quantity with the Lateral Repairs app (scan the QR codes).
- The final quality depends on the resin system used, the inversion pressure and the curing pressure. We advise against the use of other resin systems or higher pressures and accept no liability.
- All data were determined at 20 °C / 68 °F and are bench-scale values that can differ on industrial job sites.

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Technical data sheet

MULTIline FORCE UV

High-strength, reinforced hose liner for the structural rehabilitation of pipes.

Product features	
Product name	Lateral Repairs MULTIline FORCE UV
Product code	FORCEUV
Length	50 m; 100 m / 164 feet; 328 feet
Diameter	100 mm – 600 mm / 4 inch – 24 inch
Wall thickness	3,30 mm / 0,13 inch
Liner undersized	10 %
Heat resistance	Max. 80 °C / 176 °F
Negotiating bends	Max. 45°

Material features	
Material	Polyester needle-punched fleece liner with filament reinforcement and a thermoplastic polyurethane (TPU) coating.
Textile	Polyester
Textile weight	500 g/m²
Coating	TPU
Coating weight	150 g/m²
Colour / coating	White / Transparent
Water penetration	≥ 500 mbar
Storage	Protected from light, dry

Features			Length		Resin	Pressure		Roller
Dimension	Flat (mm)	Bend	L + *	L + **	kg / m	Inversion (bar)	3D (bar)	Roller gap
DN 100	~140	2 x 45	~3 cm	–	~ 1.15	0,35-0,40	–	7mm
DN 125	~175	2 x 45	~3 cm	–	~1.44	0,30-0,45	–	7mm
DN 150	~212	2 x 45	~3 cm	–	~1.72	0,30-0,50	–	7mm
DN 200	~282	2 x 45	~3 cm	–	~2.30	0,25-0,40	–	7mm
DN 225	~318	2 x 45	~3 cm	–	~2.60	0,20-0,40	–	7mm
DN 250	~353	2 x 45	~3 cm	–	~2.90	0,20-0,40	–	7mm
DN 300	~424	2 x 45	~3 cm	–	~3.21	0,20-0,40	–	7mm

Additional length *per bend >45° **Approx 4 % loss in length for every 25 % change in dimension

Resin quantity

Calculate the exact resin amount with the free Lateral Repairs app.

iPhone · App Store   Android · Google Play

Notice

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- Length addition: add the stated percentage of the liner length, plus the cm value per 45° bend; calculate the exact resin quantity with the Lateral Repairs app (scan the QR codes).
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Technical data sheet

Calibration Hose **Welded · Violet**

PVC-coated polyester fabric with an ultra-flexible, overlapped and heat-welded seam — suitable for use with most common resin types (UV Vinyl Ester, Silicate, Epoxy).

Material & construction

Material	Welded LD — Violet
Base fabric	PVC-coated polyester
Seam	Overlapped and heat-welded seam (light duty).
Compatible resins	UV Vinyl Ester · Silicate · Epoxy
Standard roll lengths	50 m, 100 m
Maximum working temperature	50 °C

Maximum recommended pressure

Pipe diameter (mm)	50	70	100	125	150	200	225	250	300
Maximum recommended pressure (Bar)	0.80	0.68	0.55	0.51	0.47	0.45	0.44	0.42	0.40

Recommended storage

Avoid extremes of temperature	• Freezing may cause the coating structure to degrade locally, especially areas where the coating is in tension or compression – at bends and edges, and immediately adjacent to seam welds. • Recommended storage temperature 5 °C – 35 °C. • Shelf life at this temperature: in excess of 1 year.
Recommended humidity	• Very high relative humidity (especially at high temperature such as tropical countries) could affect the pigmentation of the hose. • Recommended storage humidity 25 % – 65 % rh. • Shelf life at 65 %, 35 °C: 1 year.
Avoid prolonged wet storage	• As with high humidity, the coating is more susceptible to degradation at higher temperatures, and even further susceptible if the pH of liquid in contact is significantly above or below 7. • Wet storage is not recommended.
Avoid sunlight / UV	• Prolonged exposure to ultraviolet light can affect the pigmentation of the hose.

Recommended handling

Mechanical damage to be avoided. Ensure that the hose is not placed directly onto grit or gravel floor – sweep and cover the floor first. Handle the hose with care. Ensure personnel are instructed not to walk on the hose.

Further recommendations

- All hose supplied is recommended as single use only. Multiple use of the material is at customer risk.
- When in use, the hose is to be supported outside the pipe.
- Due to sizing requirements at manufacture, the customer must specify the intended use as calibration hose or pre-liner.

Notice

- The information in this data sheet corresponds to our knowledge and experience at present and is given without warranty; check the product's suitability for the intended application before use.
- All values are guideline figures determined under laboratory conditions and can differ on industrial job sites.

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Technical data sheet

Calibration Hose **Welded**

PVC-coated polyester fabric with an ultra-flexible, overlapped and heat-welded seam — suitable for use with most common resin types (UV Vinyl Ester, Silicate, Epoxy).

Material & construction

Material	Welded LD — Transparent
Base fabric	PVC-coated polyester
Seam	Overlapped and heat-welded seam (light duty).
Compatible resins	UV Vinyl Ester · Silicate · Epoxy
Standard roll lengths	50 m, 100 m
Maximum working temperature	50 °C

Maximum recommended pressure

Pipe diameter (mm)	50	70	100	125	150	200	225	250	300
Maximum recommended pressure (Bar)	0.80	0.68	0.55	0.51	0.47	0.45	0.44	0.42	0.40

Recommended storage

Avoid extremes of temperature	• Freezing may cause the coating structure to degrade locally, especially areas where the coating is in tension or compression – at bends and edges, and immediately adjacent to seam welds. • Recommended storage temperature 5 °C – 35 °C. • Shelf life at this temperature: in excess of 1 year.
Recommended humidity	• Very high relative humidity (especially at high temperature such as tropical countries) could affect the pigmentation of the hose. • Recommended storage humidity 25 % – 65 % rh. • Shelf life at 65 %, 35 °C: 1 year.
Avoid prolonged wet storage	• As with high humidity, the coating is more susceptible to degradation at higher temperatures, and even further susceptible if the pH of liquid in contact is significantly above or below 7. • Wet storage is not recommended.
Avoid sunlight / UV	• Prolonged exposure to ultraviolet light can affect the pigmentation of the hose.

Recommended handling

Mechanical damage to be avoided. Ensure that the hose is not placed directly onto grit or gravel floor – sweep and cover the floor first. Handle the hose with care. Ensure personnel are instructed not to walk on the hose.

Further recommendations

- All hose supplied is recommended as single use only. Multiple use of the material is at customer risk.
- When in use, the hose is to be supported outside the pipe.
- Due to sizing requirements at manufacture, the customer must specify the intended use as calibration hose or pre-liner.

Notice

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Technical data sheet

Calibration Hose Stitched & Welded

PVC-coated polyester fabric with an ultra-flexible, overlapped, stitched and tape-welded seam — suitable for use with most common resin types (UV Vinyl Ester, Silicate, Epoxy).

Material & construction

Material	Stitched & Welded HD — Transparent
Base fabric	PVC-coated polyester
Seam	Overlapped, stitched and tape-welded seam (heavy duty).
Compatible resins	UV Vinyl Ester · Silicate · Epoxy
Standard roll lengths	50 m, 100 m
Maximum working temperature	80 °C

Maximum recommended pressure

Pipe diameter (mm)	100	125	150	200	225	250	300
Maximum recommended pressure (Bar)	1.70	1.60	1.40	1.20	1.00	0.80	0.60

Recommended storage

Avoid extremes of temperature	• Freezing may cause the coating structure to degrade locally, especially areas where the coating is in tension or compression – at bends and edges, and immediately adjacent to seam welds. • Recommended storage temperature 5 °C – 35 °C. • Shelf life at this temperature: in excess of 1 year.
Recommended humidity	• Very high relative humidity (especially at high temperature such as tropical countries) could affect the pigmentation of the hose. • Recommended storage humidity 25 % – 65 % rh. • Shelf life at 65 %, 35 °C: 1 year.
Avoid prolonged wet storage	• As with high humidity, the coating is more susceptible to degradation at higher temperatures, and even further susceptible if the pH of liquid in contact is significantly above or below 7. • Wet storage is not recommended.
Avoid sunlight / UV	• Prolonged exposure to ultraviolet light can affect the pigmentation of the hose.

Recommended handling

Mechanical damage to be avoided. Ensure that the hose is not placed directly onto grit or gravel floor – sweep and cover the floor first. Handle the hose with care. Ensure personnel are instructed not to walk on the hose.

Further recommendations

- All hose supplied is recommended as single use only. Multiple use of the material is at customer risk.
- When in use, the hose is to be supported outside the pipe.
- Due to sizing requirements at manufacture, the customer must specify the intended use as calibration hose or pre-liner.

Notice

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- All values are guideline figures determined under laboratory conditions and can differ on industrial job sites.

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Technical data sheet

Calibration Hose Stitched & Welded

PVC-coated polyester fabric with an ultra-flexible, overlapped, stitched and tape-welded seam — suitable for use with most common resin types (UV Vinyl Ester, Silicate, Epoxy).

Material & construction

Material	Stitched & Welded HD — Orange
Base fabric	PVC-coated polyester
Seam	Overlapped, stitched and tape-welded seam (heavy duty).
Compatible resins	UV Vinyl Ester · Silicate · Epoxy
Standard roll lengths	50 m, 100 m
Maximum working temperature	80 °C

Maximum recommended pressure

Pipe diameter (mm)	100	125	150	200	225	250	300
Maximum recommended pressure (Bar)	1.70	1.60	1.40	1.20	1.00	0.80	0.60

Recommended storage

Avoid extremes of temperature	• Freezing may cause the coating structure to degrade locally, especially areas where the coating is in tension or compression – at bends and edges, and immediately adjacent to seam welds. • Recommended storage temperature 5 °C – 35 °C. • Shelf life at this temperature: in excess of 1 year.
Recommended humidity	• Very high relative humidity (especially at high temperature such as tropical countries) could affect the pigmentation of the hose. • Recommended storage humidity 25 % – 65 % rh. • Shelf life at 65 %, 35 °C: 1 year.
Avoid prolonged wet storage	• As with high humidity, the coating is more susceptible to degradation at higher temperatures, and even further susceptible if the pH of liquid in contact is significantly above or below 7. • Wet storage is not recommended.
Avoid sunlight / UV	• Prolonged exposure to ultraviolet light can affect the pigmentation of the hose.

Recommended handling

Mechanical damage to be avoided. Ensure that the hose is not placed directly onto grit or gravel floor – sweep and cover the floor first. Handle the hose with care. Ensure personnel are instructed not to walk on the hose.

Further recommendations

- All hose supplied is recommended as single use only. Multiple use of the material is at customer risk.
- When in use, the hose is to be supported outside the pipe.
- Due to sizing requirements at manufacture, the customer must specify the intended use as calibration hose or pre-liner.

Notice

- The information in this data sheet corresponds to our knowledge and experience at present and is given without warranty; check the product's suitability for the intended application before use.
- All values are guideline figures determined under laboratory conditions and can differ on industrial job sites.

Version 1 · 11.02.24

Issue: V2026.1 · 2026.06

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Technical data sheet

Glassfiber Complex 1050

E-CR glass-fibre reinforcement complex for structural pipe rehabilitation and CIPP lining.

Technical data	
Glass composition	E-CR Glass
Weight per unit area	1050 g/m² ± 8 % (deviation from nominal)
1st layer — Chopped Strand Mat	500 g/m²
2nd layer — Woven Roving	Warp 0°: 150 g/m² · Weft 90°: 410 g/m²
Bonding	Stitching
Sewing thread (polyester)	≤ 15 g/m²
Moisture content	< 0.15 %
Edges	Trimmed
Width	125 cm / 250 cm
Coupling agent	Silane
Tube diameter, internal	70 mm

Processing / Storage

The product should be conditioned for 24 hours at room temperature in the application area prior to use.

Store in its original packaging and keep dry. Storage temperature should not exceed 35 °C.

Delivery form

Rolls are packed in stretch film and supplied on pallets.

* Other roll widths, tube diameters and lengths available on request.

Resin quantity

Calculate the exact resin amount with the free Lateral Repairs app.

iPhone · App Store

Android · Google Play

Notice

- The information in this data sheet corresponds to our knowledge and experience at the present time and does not constitute legally binding assurance of properties.
- Before using the product, check its suitability for the intended application. As processing is beyond our control, responsibility rests solely with the user.

Issue: V2026.1 · 2026.06

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Technical data sheet

End Cap **Glue**

A solvent-borne, toluene-free, nearly n-hexane-free special contact adhesive for industrial and professional use.

Suitability

Flooring, shoe and leather industry, ship, boat and car-chassis building. Very well suited for joining rubber, leather, gasket, floor and wall coverings, sheets, mouldings, metals, woodpiles and different linings and insulations.

Technical data	
Application temperature	+10 °C – +40 °C
Using temperature	min. +5 °C
Temperature resistance	-40 °C – +75 °C *
Curing time	5 – 15 min (depending on conditions)
Open time	10 – 40 min
Density	0,83 g/ml
Dosage	approx. 4 m² / l
Colour	Yellowish
Storage stability	12 months · +5 – +25 °C, dry

Handling & safety	
Phase	Slightly yellowish liquid synthetic rubber solution
Tools	Brush, roller or spray gun
Packaging	Steel cans · 1, 3, 10, 20, 200, 1000 L
Cleaning	Acetone (product and tools)
Environment	Hazardous-waste disposal; cans recyclable
Fire	Highly flammable
Transport	ADR UN 1133, class 3.1
Safety	Harmful — read the SDS before use

Operating directions

Surfaces must be clean, dry and free from grease and dust; they can be coarse-ground. Apply the adhesive in a thin, smooth layer to both surfaces with a brush or roller — for spray-gun systems it can be diluted with acetone 5–20 %. Let the glue dry 15–40 minutes depending on conditions, then press the surfaces tightly together, checking there are no air bubbles. Over-dried surfaces can be reactivated with heat; if heated, press together while still warm. The bond holds immediately, with full strength developing in about two days. The dried adhesive is freeze-resistant.

* Heat resistance of the dry seam approx. +75 °C without hardener; with LR1600 hardener (3–10 %), approx. 80–90 °C.

Notice

- Contains hydroactive solvents 15–50 %, ethyl acetate 15–50 %. Highly flammable — store in well-closed cans in a cool, well-ventilated place; the directions for storing and transporting flammable liquids apply.
- Harmful. The product health & safety data sheet (SDS) must be read before use.
- The information in this data sheet corresponds to our knowledge and experience at present and is non-binding; check the product's suitability for your application before use.

Issue: V2026.1 · 2026.06

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Safety data sheet

MFE 7516 Vinyl Ester

Styrene-free vinyl ester resin — Safety Data Sheet according to Regulation (EC) No. 1272/2008.

HAZARD OVERVIEW · GHS / CLP

Warning

H315 Causes skin irritation. H317 May cause an allergic skin reaction. H319 Causes serious eye irritation.
P280 Wear protective gloves. P305+P351+P338 IF IN EYES: rinse cautiously with water for several minutes; remove contact lenses if easy; continue rinsing.

PRODUCT IDENTIFICATION

PRODUCT
MFE 7516 Vinyl Ester (styrene-free)

PRODUCT TYPE
Environmentally friendly, corrosion-resistant resin

INTENDED USE
SU3 Industrial · SU12 Manufacture of plastics · SU22 Professional uses

STABILISER
Mequinol (≥ 180 – ≤ 220 ppm)

MANUFACTURER & EMERGENCY

MANUFACTURER
Lateral Repairs UAB
Paberžių g. 5, Tauragė, LT-72328, Lithuania

CONTACT
info@lateralrepairs.com · +370 612 12882 · Office +370 698 76581

EMERGENCY TELEPHONE
+370 612 12882, or your local emergency number

DOCUMENT	REGULATION	VERSION	DATE OF ISSUE
Safety Data Sheet	(EC) No. 1272/2008	V2026.1	2026.06

FIRST AID · QUICK REFERENCE

EYE CONTACT
Rinse thoroughly with plenty of water for at least 15 minutes and consult a physician.

SKIN CONTACT
Wash off with soap and plenty of water. Consult a physician.

INHALATION
Move to fresh air and keep at rest. Get medical advice if you feel unwell.

IF SWALLOWED
Do NOT induce vomiting. Rinse mouth with water. Consult a physician.

1. Identification

Product description: MFE 7516 Vinyl Ester (styrene-free)

Intended use: Environmentally friendly, corrosion-resistant resin. SU3 — Industrial uses; SU12 — Manufacture of plastics products (compounding and conversion); SU22 — Professional uses.

Uses advised against: No information available

Manufacturer: Lateral Repairs UAB · Paberžių g. 5, Tauragė, LT-72328, Lithuania · info@lateralrepairs.com · Phone +370 612 12882 · Office +370 698 76581

Emergency telephone: +370 612 12882, or contact your local emergency telephone number

2. Hazards identification

Signal word: Warning

H315 Causes skin irritation. **H317** May cause an allergic skin reaction. **H319** Causes serious eye irritation.

P280 Wear protective gloves. **P305+P351+P338** IF IN EYES: rinse cautiously with water for several minutes; remove contact lenses if present and easy to do; continue rinsing.

Classification (EC 1272/2008): Skin irritation H315 — Category 2; Eye irritation H319 — Category 2; Skin sensitization H317 — Category 1.

Classification (67/548/EEC · 1999/45/EC): Xi Irritant — R36/38; R43.

3. Composition / information on ingredients

Type: Vinyl ester resin, styrene-free

Stabiliser: Mequinol (≥ 180 – ≤ 220 ppm)

4. First-aid measures

General advice: Consult a physician. Show this safety data sheet to the doctor in attendance.

If inhaled: Remove victim to fresh air and keep at rest in a position comfortable for breathing. If not breathing, give artificial respiration. Consult a physician.

Skin contact: Wash off with soap and plenty of water. Consult a physician.

Eye contact: Rinse thoroughly with plenty of water for at least 15 minutes and consult a physician.

If swallowed: Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Rinse mouth with water. Consult a physician.

5. Firefighting measures

Extinguishing media: Water spray, alcohol-resistant foam, dry chemical or carbon dioxide.

Special hazards: Carbon oxides, nitrogen oxides (NOx).

Advice for firefighters: Wear self-contained breathing apparatus if necessary. Use water spray to cool unopened containers.

6. Accidental release measures

Personal precautions: Wear respiratory protection. Avoid breathing vapours, mist or gas. Ensure adequate ventilation and remove all sources of ignition. Beware of vapours accumulating to form explosive concentrations in low areas.

Environmental precautions: Prevent further leakage or spillage if safe to do so. Do not let product enter drains.

Containment / cleaning up: Contain spillage, then collect with an electrically protected vacuum cleaner or by wet-brushing and place in a container for disposal according to local regulations.

7. Handling and storage

Safe handling: Avoid contact with skin and eyes; avoid inhalation of vapour or mist. Keep away from sources of ignition — no smoking. Prevent build-up of electrostatic charge.

Storage: Store in a cool place. Keep container tightly closed in a dry, well-ventilated place; reseal opened containers and keep upright. Recommended storage temperature ≤ 25 °C. Moisture- and light-sensitive.

8. Exposure controls / personal protection

Exposure limits: Contains no substances with occupational exposure limit values.

Engineering controls: Avoid contact with skin, eyes and clothing. Wash hands before breaks and immediately after handling.

Eye / face: Face shield and safety glasses tested and approved to NIOSH (US) or EN 166 (EU).

Skin: Protective gloves to EU Directive 89/686/EEC and EN 374; inspect before use.

Body: Complete chemical-protective suit, selected per concentration and amount at the workplace.

Respiratory: Where required, full-face respirator with type ABEK (EN 14387) cartridges; if the sole means of protection, use a supplied-air respirator.

9. Physical and chemical properties

Appearance	Clear liquid
Colour	Yellow, transparent liquid
Odour	Ester-like
pH	No data available
Melting / freezing point	< −60 °C
Boiling point / range	67 °C at 4,7 hPa (lit.)
Flash point	96 °C — closed cup
Vapour pressure	0,1 hPa at 20 °C
Relative density	1,1 – 1,20 g/mL at 25 °C
Partition coeff. (octanol/water)	log Pow −0,53 at 20 °C
Other properties	No data available

10. Stability and reactivity

Stability: Contains stabiliser Mequinol. May polymerise on exposure to light.

Conditions to avoid: Exposure to light.

Incompatible materials: Strong acids, strong bases, strong oxidising agents, strong reducing agents.

Hazardous decomposition: No data available.

11. Toxicological information

Acute toxicity: LD50 oral, rat: 5050 mg/kg. LD50 dermal, rabbit: > 3 000 mg/kg.

Skin / eye: Skin (rabbit): irritating (24 h). Eyes (rabbit): moderate eye irritation (24 h, Draize).

Sensitisation: May cause sensitisation by skin contact (OECD 406).

Carcinogenicity: IARC: no component ≥ 0,1 % identified as a carcinogen.

Reproductive toxicity: Rat, female, oral — effects on fertility and on embryo/fetus reported.

12. Ecological information

Toxicity to fish: Pimephales promelas (fathead minnow): 227 mg/l — 96 h (flow-through).

Persistence / degradability: Readily biodegradable — 84 % (28 d, Closed Bottle test).

PBT / vPvB: Assessment not available (chemical safety assessment not required/conducted).

13. Disposal considerations

Waste treatment: May be burned in a chemical incinerator equipped with an afterburner and scrubber. Offer surplus and non-recyclable material to a licensed disposal company.

Contaminated packaging: Dispose of as unused product.

14. Transport information

Classification: ADR/RID, IMDG and IATA: Not dangerous goods.

UN number / class / packing group: None.

Environmental hazards: IMDG marine pollutant: no.

15. Regulatory information

Labelling: According to Regulation (EC) No 1272/2008 — see Section 2.

Chemical safety assessment: A Chemical Safety Assessment has not been carried out.

Inventory: EINECS — substance included in the regulations (√).

16. Other information

H-statements: H315 Causes skin irritation. H317 May cause an allergic skin reaction. H319 Causes serious eye irritation.

R-phrases: R36/38 Irritating to eyes and skin. R43 May cause sensitisation by skin contact.

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The above information is believed to be correct but does not purport to be all-inclusive and shall be used only as a guide. It is based on the present state of our knowledge and does not represent any guarantee of the properties of the product.

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Safety data sheet

LR Silicate Resin Waterglass · Hardener

"B" component for water glass – polyisocyanate based two-component synthetic resin. The synthetic resin (components "A" + "B") is used for the lining

HAZARD OVERVIEW · GHS / CLP

Danger

H315 Causes skin irritation. H318 Causes serious eye damage.
P262 Do not get in eyes, on skin, or on clothing. P280 Wear protective gloves/protective clothing/eye protection/face protection. P303+P361+P353 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if

PRODUCT IDENTIFICATION

PRODUCT
LR Silicate Resin Waterglass Hardener

IDENTIFIED USE
"B" component for water glass – polyisocyanate based two-component synthetic resin. The synthetic resin (components "A" + "B") is used for the lining of sewer pipes and manholes. The application has to be carried out under professional, industrial conditions by persons having proper previous training.

MANUFACTURER & EMERGENCY

MANUFACTURER
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EMERGENCY
Regional Medicines and Poisons Information Centre NI, Belfast Tel.: +44 844 892 0111 (24 hrs)

DOCUMENT	REGULATION	VERSION	DATE OF ISSUE
Safety Data Sheet	1907/2006 · 2015/830	1.0 / EN	01/06/2020

1. Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

LR Silicate Resin Type Waterglass (HARDENER)

1.2. Relevant identified uses of the substance or mixture and uses advised against

"B" component for water glass – polyisocyanate based two-component synthetic resin. The synthetic resin (components "A" + "B") is used for the lining of sewer pipes and manholes. The application has to be carried out under professional, industrial conditions by persons having proper previous training.

1.3. Details of the supplier of the safety data sheet

Producer/Supplier:

UAB Lateral Repairs

Street/POB:

Paberziu g. 5

Postcode/City/Country:

LT-72328, Taurage, Lithuania

E-mail address for a competent person responsible for the safety data sheet:

info@lateralrepairs.com

Phone:

+370 698 76581

1.4. Emergency telephone number

Regional Medicines and Poisons Information Centre NI, Belfast

Tel.: +44 844 892 0111 (24 hrs)

2. Hazards identification

2.1. Classification of the substance or mixture

Classification according to Regulation (EC) No 1272/2008 (CLP):

Hazard class / category	Code	Hazard statement
Skin Irrit. 2	H315	Causes skin irritation.
Eye Dam. 1	H318	Causes serious eye damage.

2.2. Label elements

Labelling according to Regulation (EC) No 1272/2008 (CLP)

Hazard pictograms:

Signal word:

Danger

Hazard statements:

H315 Causes skin irritation.

H318 Causes serious eye damage.

Precautionary statements:

P262 Do not get in eyes, on skin, or on clothing.

P280 Wear protective gloves/protective clothing/eye protection/face protection.

P303+P361+P353 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower.

P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

Hazard determining component(s) for labelling:

Silicic acid, sodium salt

2.3. Other hazards

The substances in the mixture do not meet the PBT/vPvB criteria according to REACH, Annex XIII.

3. Composition/information on ingredients

3.1. Mixtures

Chemical characterization

Substance	EC No.	CAS No.	REACH Reg. No.	Content (%)	Classification (CLP)[1]
Silicic acid, sodium salt (Molar ratio Na_2O : SiO_2 = 1 : > 1.6 – < 2.6)	215-687-4	1344-09-8	01-2119448725-31	25–50	Skin Irrit. 2 (H315), Eye Dam. 1 (H318)
Water	231-791-2	7732-18-5	—	50–75	—

[1] – See Section 16 for the full text of the abbreviations declared above.

4. First aid measures

4.1. Description of first aid measures

General information:
No special measures necessary. When in doubt or if symptoms are observed, get medical advice. First aid responders should pay attention to self-protection.

4.1.1. Inhalation:

Remove to fresh air. Obtain medical attention if breathing difficulty persists.

4.1.2. Skin contact:

Soiled, fairly soaked clothing must be immediately removed. In case of contact with skin, wash off immediately with plenty of water. Do not allow the product to dry on the skin. Consult a doctor if skin irritation persists.

4.1.3. Eye contact:

Immediately wash affected eyes carefully, but thoroughly for at least 15 minutes under running water with eyelids held open. Use an eyewash station if possible. Consult an eye specialist.

4.1.4. Ingestion:

Immediately rinse mouth and drink plenty of water. Do not induce vomiting. Never give anything by mouth to an unconscious person or a person with cramps. Call a physician immediately.

4.2. Most important symptoms and effects, both acute and delayed

No symptoms known so far.

4.3. Indication of any immediate medical attention and special treatment needed

Hints for the physician/dangers:
The product contains alkali silicate. After swallowing and vomiting aspiration into lung is possible, which can cause chemical pneumonia or asphyxia.

5. Firefighting measures

5.1. Extinguishing media

Suitable extinguishing media:
The product itself is non-combustible. Adapt fire extinguishing measures to surrounding areas. Water spray, carbon dioxide, dry chemical extinguishing agent (powder), foam.

Non-suitable extinguishing media:
full water jet

5.2. Special hazards arising from the substance or mixture

None known.

5.3. Advice for firefighters

Special protective equipment:
Use suitable breathing apparatus. In case of fire and/or explosion do not breathe fumes.

Further information:
Fire in vicinity poses risk of pressure build-up and burst. Containers at risk from fire should be cooled with water spray. Fire residues and contaminated extinguishing water should be disposed of in accordance with local authority regulations. Carbon dioxide should be applied carefully in confined spaces since it can displace oxygen.

6. Accidental release measures

6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective clothing. Avoid contact with skin, eyes, and clothing. High risk of slipping due to leakage/spillage of product.

6.2. Environmental precautions

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Do not allow to enter drains or waterways. Prevent spread over a wide area (e.g. by containment or oil barriers).

6.3. Methods and material for containment and cleaning up

Take up with absorbent material (e.g. sand, kieselguhr, universal binder). Rinse away rest with plenty of water. Dispose according to authority regulations.

6.4. Reference to other sections

Safe handling:
see Section 7

Personal protective equipment:
see Section 8

Disposal:
see Section 13

7. Handling and storage

7.1. Precautions for safe handling

Observe the usual precautions for handling chemicals. Open and handle container with care. Avoid contact with skin, eyes and clothing. Do not breathe aerosol. Wear personal protective equipment. The product is not combustible or explosive, but heating leads to pressure build-up and risk of burst.

7.2. Conditions for safe storage, including any incompatibilities

Requirements for storage rooms and vessels: Keep only in the original container. Keep container tightly closed. Provide adequate protection against leakage, e.g. with the aid of collection trays or lowered areas. Provide a solvent-resistant and solid floor.

Further information on storage conditions:
Protect from frost.

Recommended storage temperature:
between +5 and +45 °C

Storing together hints:
Keep away from food, drink, and feeding stuff. Do not store with acids.

Storage stability:
Under correct storing conditions the product is stable for at least 12 months.

7.3. Specific end use(s)

Only for industrial users.

8. Exposure controls/personal protection

8.1. Control parameters

Limit values according to REACH registration:

DNEL for workers: 5.61 mg/m3 (inhalation); 1.59 mg/kg bw/day (dermal)

PNEC for freshwater: 7.5 mg/L, for sewage treatment plant: 348 mg/L

8.2. Exposure controls

General protective and hygiene measures:
Observe the usual precautions when handling chemicals. Avoid contact with skin and eyes. Do not breathe aerosol. Do not eat, drink or smoke during working time. Take off immediately all contaminated clothing and wash before reuse. Hands and/or face should be washed before breaks and at the end of the shift. Take a shower, if necessary. Apply skin care products after work.

Occupational exposure controls:

Respiratory protection:
Usually no personal respiratory protection is needed. It is only required when the exposure limit value is exceeded, or in the event of aerosol/mist formation. Suitable respiratory protection apparatus: P2 particle filter device (DIN EN 143).

Hand protection:
Tested protective gloves must be worn (DIN EN 374). Suitable material: NBR (nitrile rubber), butyl rubber. Thickness >= 0,4 mm, breakthrough time (maximum wearing time) >= 480 min. Observe the wear time limits as specified by the manufacturer.

Eye protection:
Safety glasses with side protection (DIN EN 166).

Body protection:
Clothing as usual in the chemical industry.

Environmental exposure controls:

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See Section 7. No additional measures necessary.

9. Physical and chemical properties

9.1. Information on basic physical and chemical properties

Appearance	liquid, clear, colourless to slightly yellow
Odour	odourless
Odour threshold	not applicable
pH-value	13-14
Melting point/freezing point	no data
Boiling range	> 100 °C
Flash point	does not ignite
Evaporation rate	no data
Flammability (solid, gaseous)	not applicable
Ignitable, explosive range	not applicable
Vapour pressure	no data
Vapour density	no data
Density	1.5 ± 0.1 g/cm3 (25 °C)
Solubility	completely miscible with water
Partition coefficient n-octanol/water	not applicable
Self-ignition temperature	not applicable
Decomposition temperature	not applicable
Dynamic viscosity	400 ± 50 mPa.s (25 °C)
Explosive properties	non-explosive
Oxidizing properties	non-oxidizing

9.2. Other information

Solids content: ca. 46%

10. Stability and reactivity

10.1. Reactivity

Under normal conditions of storage and use, hazardous reactions will not occur.

10.2. Chemical stability

Under normal conditions of storage and use, hazardous reactions will not occur.

10.3. Possibility of hazardous reactions

Under normal conditions of storage and use, hazardous reactions will not occur.

10.4. Conditions to avoid

Protect from frost.

10.5. Incompatible materials

Substances to avoid:

acids. Reacts with metals like aluminium, tin, zinc under evolution of hydrogen and heat.

10.6. Hazardous decomposition products

No hazardous decomposition products if stored and handled as prescribed/indicated (see Section 7).

11. Toxicological information

11.1. Information on toxicological effects

Acute toxicity

Based on available data, the classification criteria are not met.

Silicic acid, sodium salt (Molar ratio Na_2O : SiO_2 = 1 : > 1.6 – < 2.6) · CAS 1344-09-8

Exposure route	Dose	Species	Source
oral	LD50 > 2000 mg/kg	Rat	IUCLID
dermal	LD50 > 5000 mg/kg	Rat	IUCLID

Irritation and corrosivity	

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	Causes skin irritation. Causes serious eye damage.
Sensitizing effects	Based on available data, the classification criteria are not met.
Carcinogenic/mutagenic/toxic effects for reproduction	Based on available data, the classification criteria are not met.
STOT – single exposure	Based on available data, the classification criteria are not met.
STOT – repeated exposure	Based on available data, the classification criteria are not met.
Aspiration hazard	Based on available data, the classification criteria are not met.

12. Ecological information

The data refer to the product with the given composition and were used cross-referenced.

12.1. Toxicity

The product has not been tested.

Silicic acid, sodium salt (Molar ratio Na_2O : SiO_2 = 1 : > 1.6 – < 2.6) · CAS 1344-09-8

Aquatic toxicity	Dose	Time	Species	Source
Acute fish toxicity	LC50 1108 mg/l	96 h	Brachydanio rerio (zebrafish)	IUCLID
Acute algae toxicity	ErC50 207 mg/l	72 h	Scenedesmus subspicatus	IUCLID
Acute crustacea toxicity	EC50 1700 mg/l	48 h	Daphnia magna (big water flea)	IUCLID

12.2. Persistence and degradability

No data available.

12.3. Bioaccumulative potential

No data available.

12.4. Mobility in soil

No data available.

12.5. Results of PBT and vPvB assessment

The substances in the mixture do not meet the PBT/vPvB criteria according to REACH, Annex XIII.

12.6. Other adverse effects

The product is an alkali. Before discharge into sewage plants the product normally needs to be neutralised.

Further information:

Do not allow uncontrolled discharge of product into the environment. Do not allow to enter soil, waterways or waste water canal.

13. Disposal considerations

13.1. Waste treatment method

Advice on disposal:

Dispose according to legislation. Consult the appropriate local waste disposal expert about waste disposal.

Waste disposal number of waste from residues/unused products:

060205 WASTES FROM INORGANIC CHEMICAL PROCESSES; wastes from the MFSU of bases; other bases; hazardous waste

Contaminated packaging:

Handle contaminated packages in the same way as the substance itself. Packaging which cannot be properly cleaned must be disposed of. Non- contaminated packages may be recycled. Do not mix with other wastes.

14. Transport information

Land transport (ADR/RID) / Inland waterways transport (ADN)/ Marine transport (IMDG) / Air transport (ICAO-TI/IATA-DGR)

14.1. UN number

No dangerous good in sense of this transport regulation.

14.2. UN proper shipping name

No dangerous good in sense of this transport regulation.

14.3. Transport hazard class(es)

No dangerous good in sense of this transport regulation.

14.4. Packing group

No dangerous good in sense of this transport regulation.

14.5. Environmental hazards

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ENVIRONMENTALLY HAZARDOUS: no Danger releasing substance: No dangerous good in sense of this transport regulation.

14.6. Special precautions for user

No dangerous good in sense of this transport regulation.

14.7. Transport in bulk according to Annex II of Marpol and the IBC Code

No dangerous good in sense of this transport regulation.

15. Regulatory information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

EU regulatory information:

2010/75/EU (VOC): 0%

Information according to 2012/18/EU (SEVESO III): Not subject to 2012/18/EU (SEVESO III)

Additional information:

The product does not contain substances of very high concern (SVHC).

National regulatory information:

Water contaminating class (D): 1 – slightly water contaminating

15.2. Chemical safety assessment

In accordance with REACH chemical safety assessment has not been carried out for the product.

16. Other information

The above information describes exclusively the safety requirements of the product and is based on our present- day knowledge. The information is intended to give you advice about the safe handling of the product named in this safety data sheet, for storage, processing, transport and disposal. The information cannot be transferred to other products. In the case of mixing the product with other products or in the case of processing, the information on this safety data sheet is not necessarily valid for the new made-up material. The information is based on present level of our knowledge. It does not, however, give assurances of product properties and establishes no contract legal rights.

16.1. Indication of changes

The complete data sheet is considered new.

16.2. Abbreviations and acronyms

bw	bodyweight
CAS No.	Chemical Abstracts Service number
CLP	Regulation on Classification, Labelling and Packaging
DNEL	Derived no effect level
EC	European Commission
EC No.	EINECS and ELINCS number
EC50	Half maximal effective concentration
ErC50	EC50 based on growth rate
EINECS	European Inventory of Existing Commercial Chemical Substances
ELINCS	European List of Notified Chemical Substances
EU	European Union
LC50	Lethal concentration, 50%
LD50	Median lethal dose
MFSU	Manufacture, formulation, supply, and use
PBT	Persistent, Bioaccumulative and Toxic
PNEC	Predicted No Effect Concentration
REACH	The Registration, Evaluation, Authorisation and Restriction of Chemicals (EU regulation)
SVHC	Substances of Very High Concern
VOC	Volatile Organic Compound
vPvB	Very Persistent and Very Bioaccumulative

H-Phrases

H315	Causes skin irritation.
H318	Causes serious eye damage.

P-Phrases

P262	Do not get in eyes, on skin, or on clothing.

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P280	Wear protective gloves/protective clothing/eye protection/face protection.
P303+P361+P353	IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower.
P305+P351+P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

Hazard classes

Eye Dam.	Serious eye damage
Skin Irrit.	Skin irritation

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Safety data sheet

LR Silicate Resin Type W01 · Fast

"A" component for water glass – polyisocyanate based two-component synthetic resin. The synthetic resin (components "A" + "B") is used for the lining

HAZARD OVERVIEW · GHS / CLP


Danger

H302 Harmful if swallowed. **H315** Causes skin irritation. **H317** May cause an allergic skin reaction. **H319** Causes serious eye irritation. **H332** Harmful if inhaled. **H334** May cause allergy or asthma symptoms or breathing difficulties if inhaled. **H335** May cause respiratory irritation. **H351** Suspected of causing cancer. **H373** May cause damage to organs through prolonged or repeated exposure: respiratory system, inhalation. **P260** Do not breathe dust/fume/gas/mist/vapours/spray. **P280** Wear protective gloves/protective clothing/eye protection/face protection. **P284** Wear respiratory protection. **P302+P352** IF ON SKIN: Wash with plenty of soap and water.

PRODUCT IDENTIFICATION

PRODUCT

LR Silicate Resin Type W01 Fast

IDENTIFIED USE

"A" component for water glass – polyisocyanate based two-component synthetic resin. The synthetic resin (components "A" + "B") is used for the lining of sewer pipes and manholes. The application has to be carried out under professional or industrial conditions by persons having proper previous training.

MANUFACTURER & EMERGENCY

MANUFACTURER

UAB Lateral Repairs
Paberžių g. 5, Tauragė, LT-72328, Lithuania

CONTACT

info@lateralrepairs.com · +370 698 76581

EMERGENCY

Regional Medicines and Poisons Information Centre NI, Belfast Tel.: +44 844 892 0111 (24 hrs)

DOCUMENT	REGULATION	VERSION	DATE OF ISSUE
Safety Data Sheet	1907/2006 · 2015/830	1.0 / EN	01/06/2020

1. Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

LR Silicate Resin Type W01 (Fast)

1.2. Relevant identified uses of the substance or mixture and uses advised against

"A" component for water glass – polyisocyanate based two-component synthetic resin. The synthetic resin (components "A" + "B") is used for the lining of sewer pipes and manholes. The application has to be carried out under professional or industrial conditions by persons having proper previous training.

1.3. Details of the supplier of the safety data sheet

Producer/Supplier:

UAB Lateral Repairs

Street/POB:

Paberziu g. 5

Postcode/City/Country:

LT-72328, Taurage, Lithuania

E-mail address for a competent person responsible for the safety data sheet:

info@lateralrepairs.com

Phone:

+370 698 76581

1.4. Emergency telephone number

Regional Medicines and Poisons Information Centre NI, Belfast

Tel.: +44 844 892 0111 (24 hrs)

2. Hazards identification

2.1. Classification of the substance or mixture

Classification according to Regulation (EC) No 1272/2008 (CLP):

Hazard class / category	Code	Hazard statement
Acute Tox. 4	**H302**	Harmful if swallowed.
Skin Irrit. 2	**H315**	Causes skin irritation.
Skin Sens. 1B	**H317**	May cause an allergic skin reaction.
Eye Irrit. 2	**H319**	Causes serious eye irritation.
Acute Tox. 4	**H332**	Harmful if inhaled.
Resp. Sens. 1	**H334**	May cause allergy or asthma symptoms or breathing difficulties if inhaled.
STOT SE 3	**H335**	May cause respiratory irritation.
Carc. 2	**H351**	Suspected of causing cancer.
STOT RE 2	**H373**	May cause damage to organs through prolonged or repeated exposure: respiratory system, inhalation.

2.2. Label elements

Labelling according to Regulation (EC) No 1272/2008 (CLP)

Hazard pictograms:

Signal word:

Danger

Hazard statements:

H302 Harmful if swallowed.

H315 Causes skin irritation.

H317 May cause an allergic skin reaction.

H319 Causes serious eye irritation.

H332 Harmful if inhaled.

H334 May cause allergy or asthma symptoms or breathing difficulties if inhaled.

H335 May cause respiratory irritation.

H351 Suspected of causing cancer.

H373 May cause damage to organs through prolonged or repeated exposure: respiratory system, inhalation.

Precautionary statements:

P260 Do not breathe dust/fume/gas/mist/vapours/spray.

P280 Wear protective gloves/protective clothing/eye protection/face protection.

P284 Wear respiratory protection.

P302+P352 IF ON SKIN: Wash with plenty of soap and water.

P304+P340 IF INHALED: Remove person to fresh air and keep comfortable for breathing.

P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses if present and easy to do. Continue rinsing.

P308+P313 If exposed: Call a POISON CENTER or doctor/physician.

Hazard determining component(s) for labelling:

Isocyanic acid, polymethylenepolyphenylene ester; Tris(2-chloro-1-methylethyl) phosphate

2.3. Other hazards

The mixture does not meet persistent (P) and bioaccumulation (B) criteria, but it meets the criteria for toxicity (T). The mixture is not PBT or vPvB.

3. Composition/information on ingredients

3.1. Mixtures

Chemical characterization

Substance	EC No.	CAS No.	REACH Reg. No.	Content (%)	Classification (CLP)[1]
Isocyanic acid, polymethylene-polyphenylene ester (Polymeric MDI)[2]	(polymer)	9016-87-9	(polymer)	> 60	Acute Tox. 4 (H332), Skin Irrit. 2 (H315), Eye Irrit. 2 (H319), Resp. Sens. 1 (H334), Skin Sens. 1B (H317), Carc. 2 (H351), STOT SE 3 (H335), STOT RE 2 (H373)
Tris(2-chloro-1-methylethyl) phosphate (TCPP)	237-158-7	13674-84-5	01-2119486772-26	> 10	Acute Tox. 4 (H302)
4,4'-Methylenediphenyl diisocyanate, oligomeric reaction products with 2,4'-diisocyanatodiphenylmethane, 2,2'-methylenediphenyl diisocyanate and α-hydro-ω-hydroxypoly[oxy(methyl-1,2-ethanediyl)][3]	951-860-7	158885-25-7	(polymer)	≤ 10	Acute Tox. 4 (H332), Skin Irrit. 2 (H315), Eye Irrit. 2 (H319), Resp. Sens. 1 (H334), Skin Sens. 1B (H317), Carc. 2 (H351), STOT SE 3 (H335), STOT RE 2 (H373)
Triisobutyl phosphate	204-798-3	126-71-6	01-2119957118-32	≤ 10	Skin Sens. 1B (H317)

[1] – See Section 16 for the full text of the abbreviations declared above.

[2] – Contains < 35% 4,4'-MDI (4,4'-methylenediphenyl diisocyanate) (CAS: 101-68-8).

[3] – Contains ca. 10% 4,4'-MDI (4,4'-methylenediphenyl diisocyanate) (CAS: 101-68-8).

4. First aid measures

4.1. Description of first aid measures

General advice:

Soiled, fairly soaked clothing and shoes must be immediately removed.

4.1.1. In case of inhalation:

If inhaled, remove to fresh air. If not breathing, give artificial respiration. Get medical attention immediately.

4.1.2. In case of skin contact:

In the event of contact with the skin, at first wipe off with a paper towel/textile, then wash alternately with polyethylene glycol (if available) and water, or with plenty of warm water and soap for several minutes. Consult a doctor in the event of a skin reaction. Wash the less contaminated clothing before reuse. Clean shoes thoroughly before reuse.

4.1.3. In case of eye contact:

Hold the eyes open and rinse with water for a sufficiently long period of time (at least 10 minutes). Get medical attention immediately.

4.1.4. In case of ingestion:

DO NOT induce the patient to vomit, medical advice is required. Never give anything by mouth to an unconscious person. Provided the patient is conscious, wash out mouth with water.

4.1.5. Information to physician:

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The product irritates the respiratory tract and may trigger sensitisation of the skin and respiratory tract. Treatment of acute irritation or bronchial constriction is primarily symptomatic. Following severe exposure the patient should be kept under medical review for at least 48 hours.

4.2. Most important symptoms and effects, both acute and delayed

Headache, nausea, shortness of breath, sore throat, redness on the skin. Repeated or prolonged contact may cause skin sensitization. Repeated or prolonged inhalation exposure may cause asthma.

4.3. Indication of any immediate medical attention and special treatment needed

Depending on the degree of exposure, periodic medical examination is suggested.

5. Firefighting measures

5.1. Extinguishing media

Suitable extinguishing media:

Foam, CO_2 or dry powder. Water spray may be used if no other available and then in copious quantities.

Unsuitable extinguishing media:

High volume water jet.

5.2. Special hazards arising from the substance or mixture

Carbon dioxide, carbon monoxide, hydrogen cyanide, nitrogen oxides, isocyanate vapours. The substances/groups of substances mentioned can be released in case of fire.

5.3. Advice for firefighters

Reaction between water and hot isocyanate may be vigorous (strongly exothermic). Prevent washings from entering watercourses. Keep fire-exposed containers cool by spraying with water.

Special protective equipment:

Fire-fighters should wear appropriate protective equipment and self- contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode. Safety boots, gloves, safety helmet and protective clothing should be worn.

Further information:

In the event of fire and/or explosion do not breathe fumes. Fire in vicinity poses risk of pressure build-up and rupture. Containers at risk from fire should be cooled with water and, if possible, removed from the danger area. Due to reaction with water producing CO_2 gas, a hazardous build-up of pressure could result if contaminated containers are re-sealed. Containers may burst if overheated. Do not allow contaminated extinguishing water to enter the soil, groundwater or surface waters.

6. Accidental release measures

6.1. Personal precautions, protective equipment and emergency procedures

Immediately contact emergency personnel. Evacuate the area. Keep upwind to avoid inhalation of vapours. Clean-up should only be performed by trained personnel. Keep unauthorized persons away.

6.1.1. For non-emergency personnel:

Remove not affected people. Inform the relevant emergency services and authorities.

6.1.2. For emergency responders:

People dealing with major spillages should wear full protective clothing including respiratory protection. Use suitable protective equipment.

6.2. Environmental precautions

Do not allow contaminated extinguishing water to enter the soil, groundwater or surface waters. Avoid dispersal of spilt material and runoff and contact with drains and sewers.

6.3. Methods and material for containment and cleaning up

Absorb spillages onto sand, earth or any suitable adsorbent material. Leave to react for at least 30 minutes. Do not absorb onto sawdust or other combustible materials. Contaminated adsorbent material shall be disposed according to Section 13. Wash the spillage area with water.

6.4. Reference to other sections

Information regarding exposure controls/personal protection and disposal considerations can be found in Section 8 and 13.

7. Handling and storage

7.1. Precautions for safe handling

7.1.1. Protective measures:

Provide sufficient air exchange and/or exhaust in work rooms. In all workplaces of the plant where high concentrations of isocyanate aerosols and/or vapours may be generated (e.g. during pressure release, mould venting or when cleaning mixing heads with an air blast), appropriately located exhaust ventilation must be provided in order to prevent occupational exposure limits from being exceeded. The air should be drawn away from the personnel handling the product. The efficiency of the ventilation system must be monitored regularly because of the possibility

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of blockage. Atmospheric concentrations should be minimised and kept as low as reasonably practicable below the occupational exposure limit.

7.1.2. Advice on general occupational hygiene:

No eating, drinking, smoking or tobacco use at the place of work.

Contact with skin and eyes and inhalation of vapours must be avoided under all circumstances. Keep equipment clean. A basic essential in sampling, handling and storage is the prevention of contact with water. Keep stocks of decontaminant readily available.

7.2. Conditions for safe storage, including any incompatibilities

Store and transport in separate, airtight vessels, between +10 ˚C and +25 ˚C. The containers and vessels shall be protected from direct sunshine and other weather impacts. Keep container tightly closed and sealed until ready for use. Containers that have been opened must be carefully resealed and kept upright to prevent leakage. Do not store in unlabelled containers. Use appropriate container to avoid environmental contamination.

7.3. Specific end use(s)

For the relevant identified use(s) listed in Section 1 the advice mentioned in this section is to be observed.

8. Exposure controls/personal protection

8.1. Control parameters

8.1.1. Occupational exposure limits in air

A workplace exposure limit (WEL) of 0.02 mg/m3 for total isocyanates (as NCO) as an 8-hour TWA, and a short term WEL (15 min) of 0.07 mg/m3 have been assigned in the United Kingdom. A BMGV for isocyanates, based on the measurement of urinary diamines, has been set at 1 µmol diamine/mol creatinine.

8.1.2. DNEL/PNEC-values

The risk characterization of PMDI (CAS: 9016-87-9) is the following:

Workers:

Acute/short-term exposure – systemic effects (dermal): DNEL = 50 mg/kg bw/day

Acute/short-term exposure – systemic effects (inhalation): DNEL = 0.1 mg/m^3

Acute/short-term exposure – local effects (dermal): DNEL = 28.7 mg/cm2

Acute/short-term exposure – local effects (inhalation): DNEL = 0.1 mg/m^3

Long-term exposure – systemic effects (inhalation): DNEL = 0.05 mg/m^3

Long-term exposure – systemic effects (dermal): Not applicable.

Long-term exposure – local effects (inhalation): DNEL = 0.05 mg/m^3

Long-term exposure – local effects (dermal): Not applicable.

PNEC sediment: As PMDI is a reactant with water, access of water to PMDI and vice versa is strictly controlled. Furthermore, PMDI polymerizes in the presence of water and thus exposure of PMDI to sediment is highly likely to be negligible. Therefore, PNEC sediment cannot be derived for PMDI.

PNEC soil: 1 mg/kg soil dw

PNEC oral: There are no data on effects of oral PMDI to birds. Exposure to birds is not expected and data from experimental animals show PMDI to be of low oral toxicity.

8.2. Exposure controls

Respiratory protection:
Respiratory protection in case of vapour/aerosol release. Combination filter for organic, inorganic, acid inorganic, and basic gases/vapours (e.g. EN 14387 Type ABEK) shall be used.

Hand protection:
Chemical resistant protective gloves (EN 374)

Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN 374):
butyl rubber (butyl) – 0.7 mm coating thickness

nitrile rubber (NBR) – 0.4 mm coating thickness

chloroprene rubber (CR) – 0.5 mm coating thickness

Unsuitable materials:
polyvinyl chloride (PVC) – 0.7 mm coating thickness

polyethylene (PE) laminate – ca. 0.1 mm coating thickness

Eye protection:
Safety glasses with side shields (frame goggles) (e.g. EN 166).

Body protection:

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Safety shoes (e.g. according to EN 20346) and closed workwear.

General safety and hygiene measures:

Do not breathe vapour/spray. With products freshly manufactured from isocyanates body protection and chemical resistant protective gloves is recommended. Wearing of closed workwear is required additionally to the personal protective equipment. No eating, drinking, smoking or tobacco use at the place of work. Take off immediately all contaminated clothing. Hands and/or face should be washed before breaks and at the end of the shift. After work the skin should be cleaned and skin-care agents applied.

9. Physical and chemical properties

9.1. Information on basic physical and chemical properties

Appearance	liquid, dark-brown
Odour	damp
Odour threshold	not known
pH-value	not applicable (reacts with water)
Melting point/freezing point	not defined (mixture)
Boiling range	> 200 °C
Flash point	> 200 °C (MDI)
Evaporation rate	not defined (mixture)
Flammability (solid, gaseous)	not applicable (liquid)
Ignitable, explosive range	not defined (mixture)
Vapour pressure	< 0.00001 mbar (at 20 °C)
Vapour density	not defined (mixture)
Density	1.19 ± 0.01 g/cm3 (at 25 °C)
Solubility	reacts with water with slow CO_2 appearance at the border area into non-soluble, high melting point or not melting polyurea
Partition coefficient n-octanol/water	not applicable (mixture)
Self-ignition temperature	4,4'-MDI does not ignite till 601 °C
Decomposition temperature	not applicable (mixture)
Viscosity	160–220 mPas (at 25 °C)
Explosive properties	non-explosive
Oxidising properties	non-oxidizing

9.2. Other information

No data.

10. Stability and reactivity

10.1. Reactivity

Reacts with water, acids, alcohols, amines, bases, and oxidants.

10.2. Chemical stability

The main removal mechanism of MDIs in the environment is hydrolysis. MDI reacts quickly with water to form predominantly solid, insoluble polyureas. Under conditions typical of many types of environmental contact, i.e. with relatively poor dispersion of the isocyanate, the interfacial reaction leads to the formation of a solid crust encasing partially reacted product. This crust restricts ingress of water and egress of amine, and hence slows and modifies hydrolysis.

Stability in organic solvents:

All MDI isomers and forms are highly unstable in dimethyl sulphoxide (DMSO) solvent, water content of the DMSO is increasing breakdown. MDI is more stable in EGDE (ethylene glycol dimethyl ether) solvent.

(Read-across based on 4,4'-methylenediphenyl diisocyanate – CAS 101-68-8.)

10.3. Possibility of hazardous reactions

Reaction is slow with cold or warm water (< 50 °C), with hot water or steam the reaction is faster, producing carbon-dioxide causing pressure increase. Acids, alcohols, amines, bases, and oxidants may cause fire and explosion hazard.

10.4. Conditions to avoid

High temperature, moisture, strong light.

10.5. Incompatible materials

Substances to avoid:

water, acids, alkalis, alcohols, amines.

10.6. Hazardous decomposition products

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No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological information

The mixture has not been tested. Information is related to 4,4'-Methylenediphenyl diisocyanate if no other is mentioned.

11.1. Information on toxicological effects

Acute toxicity – oral:	Harmful
Rats (female)	LD50 = 632 mg/kg Tris (2-chloro-1-methylethyl) phosphate CAS 13674-84-5
Acute toxicity – inhalation (aerosol):	Harmful
Rats	LC50 = 0.49 mg/l air (4 h) OECD Guideline 403
Rats	LC50 > 7 mg/l air (4 h), dusts and mists OECD 403 Acute Inhalation Toxicity / 433 Acute Inhalation Toxicity: Fixed Concentration Procedure Tris(2-chloro-1-methylethyl) phosphate CAS 13674-84-5
Rats	LC50 > 5.14 g/m3 (4 h), dusts/mists OECD 403 Acute Inhalation Toxicity Triisobutyl phosphate (CAS: 126-71-6)
Acute toxicity – dermal:	Not classified. Based on available data, the classification criteria are not met.
Rabbit	LD50 > 9400 mg/kg bw (24 h) OECD Guideline 402

11.2. Irritation/Corrosion

Summarized the results of the studies together with human occupational case reports support the official classification.

Skin corrosion/Skin irritation:	Irritating Irritating in rabbits. (4 h / 14 days) OECD Guideline 404

Ingredient name	Result	Species	Score	Exposure	Test
Triisobutyl phosphate (CAS: 126-71-6)	Skin erythema/ eschar	Rabbit	0.67	–	OECD 404 Acute Dermal Irritation/ Corrosion

Eye damage/Irritation:	Not irritating in rabbits. (24 h / 21 days) OECD Guideline 405

(Read-across based on methylenediphenyl diisocyanate, isomer mixture – CAS 26447-40-5.) Summarized the available animal data would not support classification of MDI as an eye irritant. But together with human occupational case reports in which symptoms of eye irritation were reported the legal classification as eye irritant should be applied.

11.3. Sensitisation

Animal data as well as studies in humans provide evidence of possible skin sensitisation, and of respiratory sensitisation due to MDI. Animal studies indicate that MDI is a strong allergen. Human case reports describe the occurrence of allergic contact dermatitis due to MDI exposure.

Skin sensitisation:	Mice Sensitizing OECD Guideline 429 (LLNA)

Ingredient name	Route of exposure	Species	Result	Test description
Triisobutyl phosphate (CAS: 126-71-6)	Skin	Guinea pig	Sensitizing	OECD 406 Skin Sens.

Respiratory sensitisation:	Rats (male) Sensitizing OECD Guideline 39

11.4. Germ cell mutagenicity

Not classified. Based on available data, the classification criteria are not met.

11.5. Carcinogenicity

Carc. 2

Rats (inhalation, aerosol):	NOAEC = 0.2 mg/m³ air (toxicity) (2 years; 6 h/day, 5 days/week) NOAEC = 1 mg/m³ air (carcinogenicity) (2 years; 6 h/day, 5 days/week) LOAEC = 6 mg/m³ air (carcinogenicity) (2 years; 6 h/day, 5 days/week) OECD Guideline 453

11.6. Reproductive toxicity

Not classified. Based on available data, the classification criteria are not met.

Effects on fertility:	No fertility, nor multigeneration studies are available.
Rats (inhalation):	NOAEL = 4 mg/m³ air (developmental toxicity) (10 days; 1/day, 6 h) NOAEL = 4 mg/m³ air (maternal toxicity) (10 days; 1/day, 6 h) OECD Guideline 414

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11.7. STOT – single exposure

MDI is irritant to the respiratory tract.

11.8. STOT – repeated exposure

Harmful

Rats (inhalation, aerosol):	NOAEC = 0.2 mg/m3 air (2 years; 6 h/day, 5 days/week) LOAEC = 1.0 mg/m3 air (2 years; 6 h/day, 5 days/week) Target organs: respiratory – lung OECD Guideline 453

11.9. Aspiration hazard

Not classified due to lack of data.

12. Ecological information

The mixture has not been tested. Information is related to 4,4'-methylenediphenyl diisocyanate if no other is mentioned.

12.1. Toxicity

12.1.1. Aquatic toxicity

Short-term toxicity to fish

Freshwater fish (Danio rerio): LC50 > 1000 mg/l (96 h)

OECD Guideline 203

Danio rerio (zebrafish): LC50 = 56.2 mg/l (96 h)

Tris(2-chloro-1-methylethyl) phosphate, CAS: 13674-84-5

Oncorhynchus mykiss (rainbow trout): LC50 = 1.6 mg/l (96 h)

Fish: LC50 = 17.8–21.5 mg/l (96 h)

Triisobutyl phosphate, CAS: 126-71-6

Long-term toxicity to fish

Data waiving. In accordance with column 2 of REACH Annex IX the long-term toxicity testing on fish shall be proposed if the chemical safety assessment according to Annex I indicates the need to investigate further the effects on aquatic organisms. The corresponding PEC/PNEC ratios would be less than 1. Taking into account the scientific and exposure arguments, it appears appropriate to waiver the long-term fish/plant/soil and sediment toxicity studies.

Short-term toxicity to aquatic invertebrates

Freshwater invertebrates (Daphnia magna): EC50 > 1000 mg/l (24 h)

OECD Guideline 202

Daphnia magna: EC50 = 131 mg/l (48 h)

Tris(2-chloro-1-methylethyl) phosphate, CAS: 13674-84-5

Daphnia – Daphnia magna Acute EC50 = 11 mg/l (48 h)

DIN 38412, Part 11

Triisobutyl phosphate, CAS: 126-71-6

Long-term toxicity to aquatic invertebrates

Freshwater invertebrates (Daphnia magna): NOEC >= 10 mg/l (21 days)

OECD Guideline 211

Toxicity to aquatic algae and cyanobacteria

Freshwater algae (Desmodesmus subspicatus) EC50 > 1640 mg/l (72 h)

OECD Guideline 201

Freshwater algae (Desmodesmus subspicatus) EC50 = 82 mg/l (72 h)

Tris(2-chloro-1-methylethyl) phosphate, CAS: 13674-84-5

Algae (Desmodesmus subspicatus) Acute IC50 = 34.1 mg/l (72 h) growth rate

DIN 3812, Part 9

Triisobutyl phosphate, CAS: 126-71-6

Algae (Desmodesmus subspicatus): Acute IC50 = 33.2 mg/l (72 h) growth rate, biomass

DIN 3812, Part 9

Triisobutyl phosphate, CAS: 126-71-6

Bacteria – Activated sludge Chronic EC50 = 37.2 mg/l (28 days)

OECD 301B Ready Biodegradability – CO_2 Evolution Test

Triisobutyl phosphate, CAS: 126-71-6

Toxicity to aquatic plants other than algae

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Data waiving. Not required by REACH annexes. However, a (soil) mesocosm study with PMDI exists in which the toxicity towards macrophytes (Potamogeton crispus and Zannichellia palustris) was assessed. No toxicity was observed at a loading of 1000 and 10000 mg/l, approximately 100% of the substance was found in the sediment as hardened material.

Toxicity to microorganisms

Microorganisms (activated sludge) EC50 > 100 mg/l (3 h)

OECD Guideline 209

Toxicity to other aquatic organisms

This information is not available, but not required under REACH.

12.1.2. Sediment toxicity

Data waiving. Annex X states that this study need not be conducted if the chemical safety assessment does not indicate a need to further investigate the effects on sediment organisms.

12.1.3. Terrestrial toxicity

Toxicity to soil macroorganisms except arthropods

Eisenia fetida LC50 > 1000 mg/kg soil dw (14 days)

OECD Guideline 207

Toxicity to terrestrial arthropods

Data waiving. Based on the chemical safety assessment and the risk assessment, there is no need to further investigate the terrestrial arthropods toxicity as there is no risk for the terrestrial environment as indicated by the PEC/PNEC ratio being < 0.239. Direct/indirect exposure to soil is unlikely.

Toxicity to terrestrial plants

Avena sativa EC50 > 1000 mg/kg soil dw (14 days)

Lactuca sativa EC50 > 1000 mg/kg soil dw (14 days)

OECD Guideline 208

Toxicity to soil microorganisms

Data waiving. Annex X states that this study need not be conducted if the chemical safety assessment does not indicate a need to further investigate the effects on sediment organisms.

Toxicity to other above-ground organisms

Data waiving. Not required by REACH annexes.

12.1.4. Conclusion on classification

Hazardous to the aquatic environment (acute)

Based on available data, the classification criteria are not met.

(EC/LC50 for fish, invertebrates and algae > 1000 mg/l)

Hazardous to the aquatic environment (chronic)

Based on available data, the classification criteria are not met.

(NOEC for algae > 1640 mg/L; NOEC for invertebrates > 10 mg/l)

12.2. Persistence and degradability

Phototransformation in air

Half-life (DT50): 0.92 days

Hydrolysis

MDI reacts with water to form predominantly inert polyurea.

Half-life (DT50): ca. 20 h (at 25 °C)

(Read-across based on oligomeric MDI – CAS 32055-14-4)

Phototransformation in water and soil

No data is available.

Biodegradation in water

Under test conditions no biodegradation was observed. (28 days)

OECD Guideline 302C

Biodegradation in water and sediment

Data waiving. In accordance with Annex XI, simulation biodegradation tests are technically not feasible as the test substance reacts quickly with water. The corresponding PEC/PNEC ratios would be less than 1. Taking into account the scientific and exposure arguments, it appears appropriate to waiver the long-term fish/plant/soil and sediment toxicity studies.

Biodegradation in soil

Data waiving. See at Biodegradation in water and sediment.

12.3. Bioaccumulative potential

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Bioaccumulation - aquatic/sediment: Due to the high reactivity of the substances of the MDI category with water, bioaccumulation tests can in principle not be performed with these substances. However, one bioaccumulation test with 4,4'-MDI and a mesocosm study with PMDI with an indication of bioaccumulation potential have been performed. As no analytical measurements were done, it cannot be determined if the values are truly related to MDI. However, based on the available information and the reactivity of MDI substances of the category approach, no new bioaccumulation study is deemed necessary.

BCF (Cyprinus carpio): 200 (28 days)

Method: OECD Guideline 305E

Terrestrial bioaccumulation

No data is available on terrestrial bioaccumulation, but it is not required under REACH.

12.4. Mobility in soil

Adsorption/desorption

Data waiving. According to Annex VIII the study need not be done if the test substance degrades rapidly. The corresponding PEC/PNEC ratios would be less than 1. Taking into account the scientific and exposure arguments, it appears appropriate to waiver the long-term fish/plant/soil and sediment toxicity studies.

Volatilisation

The Henry's Law Constant, estimated from the measured vapour pressure and the calculated water solubility, is 2.263 x 10-7 atm-m3/mole. Hence, volatilization is unlikely to be a significant removal mechanism for MDI substances of the category approach.

12.5. Results of PBT and vPvB assessment

Conclusion for the P criterion

The results from the biodegradation test indicate that PMDI is not biodegradable. Based on experimental hydrolysis and indirect photolysis half-lives, PMDI is not considered to be persistent in the environment and is identified as not P. Based on the justification in the category approach, it is assumed that all MDI analogues included in the category are not P.

Conclusion for the B criterion

Although MDI has a high measured log Pow value (4.51), a full bioaccumulation test with 4,4'-MDI indicated that the bioaccumulation potential is low. Due to the fast hydrolysis, exposure of the environment to the substance is unlikely or very low, there is no potential for significant bioaccumulation possible. Hence, 4,4'-MDI does not fulfil the requirements for the B criterion and is not identified as B. Based on the justification in the category approach, it is assumed that all MDI analogues included in the category are not B.

Conclusion for the T criterion

The concentrations tested were far above the water solubility of the MDI substances (7.5 mg/l). However, the water solubility limit of MDI is far above the criteria for T and on the basis of aquatic toxicity tests MDI is identified as not T. However, according to Annex I of 67/548/EEC MDI is classified as Xn, R48, which automatically triggers a T. Based on this classification MDI is identified as T.

12.6. Other adverse effects

It is not expected that substance has an effect on global warming, ozone depletion in the stratosphere or ozone formation in the troposphere.

Secondary poisoning

Based on the available information, there is no indication of a bioaccumulation potential and, hence, secondary poisoning is not considered relevant. Exposure to birds is not expected and data from experimental animals show MDI to be of low oral toxicity.

13. Disposal considerations

13.1. Waste treatment methods

The products becoming useless and the contaminated containers not suitable for product storage must be handled as hazardous waste in accordance with EU and regional hazardous waste regulations. European Waste Catalogue code: 08 05 01

13.1.1. Product / Packaging disposal:

Contaminated packaging should be emptied as far as possible; then it can be passed on for recycling after being thoroughly cleaned. Wrappings cleaned from contamination with suitable cleaning process (e.g. by steaming, treating with washing fluid, etc.) must be considered as non-hazardous waste.

13.1.2. Waste treatment options:

Incinerate in suitable incineration plant, observing local authority regulations.

14. Transport information

Land transport (ADR/RID/GGVSE)

Sea transport (IMDG Code/GGVSee)

Air transport (ICAO-IATA/DGR)

14.1. UN number

Not dangerous goods

14.2. UN proper shipping name

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Not dangerous goods

14.3. Transport hazard class(es)

Not dangerous goods

14.4. Packing group

Not dangerous goods

14.5. Environmental hazards

Marine pollutant: no

14.6. Special precautions for users

EmS number: Not dangerous goods

14.7. Transport in bulk according to Annex II of Marpol and the IBC Code

Not relevant

15. Regulatory information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

15.1.1. Information regarding relevant EU safety, health and environmental provisions

ISOPA, the European Diisocyanate & Polyol Producers Association has elaborated a Guideline document for the safe treatment of MDI containing products. The Guidelines have been built into this data sheet.

15.2. Chemical safety assessment

In accordance with REACH chemical safety assessment (CSA) has not been carried out for the product. However, the results from the CSA for 4,4'-MDI were transposed into this SDS.

16. Other information

The information given corresponds with our actual knowledge and experience. This information is meant to describe our product in view of possible safety requirements. Classification of the mixture is based on the classification of components.

16.1. Indication of changes

This is the first edition of the SDS.

16.2. Abbreviations and acronyms

BCF	Bioconcentration factor
BMGV	Biological monitoring guidance value
bw	bodyweight
CAS No.	Chemical Abstracts Service number
CLP	Regulation on classification, labelling and packaging
DNEL	Derived no effect level
dw	dry weight
EC No.	EINECS and ELINCS number
EC50	Half maximal effective concentration
EEC	European Economic Community
EINECS	European Inventory of Existing Commercial Chemical Substances
ELINCS	European List of Notified Chemical Substances
EU	European Union
IC50	Half maximal inhibitory concentration
LC50	Lethal concentration, 50%
LD50	Median lethal dose
LLNA	Local lymph node assay
LOAEC	Lowest Observed Adverse Effect Concentration
NOAEC	No Observed Adverse Effect Concentration
NOAEL	No Observed Adverse Effect Level
NOEC	No Observed Effect Concentration
OECD	Organisation for Economic Cooperation and Development
PBT	Persistent, Bioaccumulative and Toxic
PEC	Predicted Environmental Concentration
PMDI	Polymeric MDI (CAS: 9016-87-9)
PNEC	Predicted No Effect Concentration

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REACH	The Registration, Evaluation, Authorisation and Restriction of Chemicals
SDS	Safety Data Sheet
TWA	Time-weighted average
vPvB	Very Persistent and Very Bioaccumulative
WEL	Workplace exposure limit

16.3. Key literature references and sources for data

Safety data sheets, received from the raw materials suppliers.

16.4. Full text of abbreviations

H-Phrases

H302	Harmful if swallowed.
H315	Causes skin irritation.
H317	May cause an allergic skin reaction.
H319	Causes serious eye irritation.
H332	Harmful if inhaled.
H334	May cause allergy or asthma symptoms or breathing difficulties if inhaled.
H335	May cause respiratory irritation.
H351	Suspected of causing cancer.
H373	May cause damage to organs through prolonged or repeated exposure: respiratory system, inhalation.

P-Phrases

P260	Do not breathe dust/fume/gas/mist/vapours/spray.
P280	Wear protective gloves/protective clothing/eye protection/face protection.
P284	Wear respiratory protection.
P302+P352	IF ON SKIN: Wash with plenty of soap and water.
P304+P340	IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P305+P351+P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses if present and easy to do. Continue rinsing.
P308+P313	If exposed: Call a POISON CENTER or doctor/physician.

Hazard classes

Acute Tox.	Acute toxicity
Carc.	Carcinogenicity
Eye Irrit.	Serious eye irritation
Resp. Sens.	Respiratory sensitization
Skin Irrit.	Skin irritation
Skin Sens.	Skin sensitization
STOT RE	Specific target organ toxicity – repeated exposure
STOT SE	Specific target organ toxicity – single exposure

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LR Silicate Resin Type W · Winter

"A" component for water glass – polyisocyanate based two-component synthetic resin. The synthetic resin (components "A" + "B") is used for the lining

HAZARD OVERVIEW · GHS / CLP

Danger

H302 Harmful if swallowed. H315 Causes skin irritation. H317 May cause an allergic skin reaction. H319 Causes serious eye irritation. H332 Harmful if inhaled. H334 May cause allergy or asthma symptoms or breathing difficulties if inhaled. H335 May cause respiratory irritation. H351 Suspected of causing cancer. H373 May cause damage to organs through prolonged or repeated exposure: respiratory system, inhalation. P260 Do not breathe dust/fume/gas/mist/vapours/spray. P280 Wear protective gloves/protective clothing/eye protection/face protection. P284 Wear respiratory protection. P302+P352 IF ON SKIN: Wash with plenty of soap and water.

PRODUCT IDENTIFICATION

PRODUCT

LR Silicate Resin Type W Winter

IDENTIFIED USE

"A" component for water glass – polyisocyanate based two-component synthetic resin. The synthetic resin (components "A" + "B") is used for the lining of sewer pipes and manholes. The application has to be carried out under professional or industrial conditions by persons having proper previous training.

MANUFACTURER & EMERGENCY

MANUFACTURER

UAB Lateral Repairs
Paberžių g. 5, Tauragė, LT-72328, Lithuania

CONTACT

info@lateralrepairs.com · +370 698 76581

EMERGENCY

Regional Medicines and Poisons Information Centre NI, Belfast Tel.: +44 844 892 0111 (24 hrs)

DOCUMENT	REGULATION	VERSION	DATE OF ISSUE
Safety Data Sheet	1907/2006 · 2015/830	1.0 / EN	01/06/2020

1. Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

LR Silicate Resin Type W (Winter)

1.2. Relevant identified uses of the substance or mixture and uses advised against

"A" component for water glass – polyisocyanate based two-component synthetic resin. The synthetic resin (components "A" + "B") is used for the lining of sewer pipes and manholes. The application has to be carried out under professional or industrial conditions by persons having proper previous training.

1.3. Details of the supplier of the safety data sheet

Producer/Supplier:

UAB Lateral Repairs

Street/POB:

Paberziu g. 5

Postcode/City/Country:

LT-72328, Taurage, Lithuania

E-mail address for a competent person responsible for the safety data sheet:

info@lateralrepairs.com

Phone:

+370 698 76581

1.4. Emergency telephone number

Regional Medicines and Poisons Information Centre NI, Belfast

Tel.: +44 844 892 0111 (24 hrs)

2. Hazards identification

2.1. Classification of the substance or mixture

Classification according to Regulation (EC) No 1272/2008 (CLP):

Hazard class / category	Code	Hazard statement
Acute Tox. 4	H302	Harmful if swallowed.
Skin Irrit. 2	H315	Causes skin irritation.
Skin Sens. 1B	H317	May cause an allergic skin reaction.
Eye Irrit. 2	H319	Causes serious eye irritation.
Acute Tox. 4	H332	Harmful if inhaled.
Resp. Sens. 1	H334	May cause allergy or asthma symptoms or breathing difficulties if inhaled.
STOT SE 3	H335	May cause respiratory irritation.
Carc. 2	H351	Suspected of causing cancer.
STOT RE 2	H373	May cause damage to organs through prolonged or repeated exposure: respiratory system, inhalation.

2.2. Label elements

Labelling according to Regulation (EC) No 1272/2008 (CLP)

Hazard pictograms:

Signal word:

Danger

Hazard statements:

H302 Harmful if swallowed.

H315 Causes skin irritation.

H317 May cause an allergic skin reaction.

H319 Causes serious eye irritation.

H332 Harmful if inhaled.

H334 May cause allergy or asthma symptoms or breathing difficulties if inhaled.

H335 May cause respiratory irritation.

H351 Suspected of causing cancer.
H373 May cause damage to organs through prolonged or repeated exposure: respiratory system, inhalation.

Precautionary statements:
P260 Do not breathe dust/fume/gas/mist/vapours/spray.
P280 Wear protective gloves/protective clothing/eye protection/face protection.
P284 Wear respiratory protection.
P302+P352 IF ON SKIN: Wash with plenty of soap and water.
P304+P340 IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses if present and easy to do. Continue rinsing.
P308+P313 If exposed: Call a POISON CENTER or doctor/physician.

Hazard determining component(s) for labelling:
Isocyanic acid, polymethylenepolyphenylene ester; Tris(2-chloro-1-methylethyl) phosphate

2.3. Other hazards

The mixture does not meet persistent (P) and bioaccumulation (B) criteria, but it meets the criteria for toxicity (T). The mixture is not PBT or vPvB.

3. Composition/information on ingredients

3.1. Mixtures

Chemical characterization

Substance	EC No.	CAS No.	REACH Reg. No.	Content (%)	Classification (CLP)[1]
Isocyanic acid, polymethylene-polyphenylene ester (Polymeric MDI)[2]	(polymer)	9016-87-9	(polymer)	> 60	Acute Tox. 4 (H332), Skin Irrit. 2 (H315), Eye Irrit. 2 (H319), Resp. Sens. 1 (H334), Skin Sens. 1B (H317), Carc. 2 (H351), STOT SE 3 (H335), STOT RE 2 (H373)
Tris(2-chloro-1-methylethyl) phosphate (TCPP)	237-158-7	13674-84-5	01-2119486772-26	> 10	Acute Tox. 4 (H302)
4,4'-Methylenediphenyl diisocyanate, oligomeric reaction products with 2,4'-diisocyanatodiphenylmethane, 2,2'-methylenediphenyl diisocyanate and α-hydro-ω-hydroxypoly[oxy(methyl-1,2-ethanediyl)][3]	951-860-7	158885-25-7	(polymer)	≤ 5	Acute Tox. 4 (H332), Skin Irrit. 2 (H315), Eye Irrit. 2 (H319), Resp. Sens. 1 (H334), Skin Sens. 1B (H317), Carc. 2 (H351), STOT SE 3 (H335), STOT RE 2 (H373)

[1] – See Section 16 for the full text of the abbreviations declared above.

[2] – Contains < 35% 4,4'-MDI (4,4'-methylenediphenyl diisocyanate) (CAS: 101-68-8).

[3] – Contains < 10% 4,4'-MDI (4,4'-methylenediphenyl diisocyanate) (CAS: 101-68-8).

4. First aid measures

4.1. Description of first aid measures

General advice:
Soiled, fairly soaked clothing and shoes must be immediately removed.

4.1.1. In case of inhalation:

If inhaled, remove to fresh air. If not breathing, give artificial respiration. Get medical attention immediately.

4.1.2. In case of skin contact:

In the event of contact with the skin, at first wipe off with a paper towel/textile, then wash alternately with polyethylene glycol (if available) and water, or with plenty of warm water and soap for several minutes. Consult a doctor in the event of a skin reaction. Wash the less contaminated clothing before reuse. Clean shoes thoroughly before reuse.

4.1.3. In case of eye contact:

Hold the eyes open and rinse with water for a sufficiently long period of time (at least 10 minutes). Get medical attention immediately.

4.1.4. In case of ingestion:

DO NOT induce the patient to vomit, medical advice is required. Never give anything by mouth to an unconscious person. Provided the patient is conscious, wash out mouth with water.

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4.1.5. Information to physician:

The product irritates the respiratory tract and may trigger sensitisation of the skin and respiratory tract. Treatment of acute irritation or bronchial constriction is primarily symptomatic. Following severe exposure the patient should be kept under medical review for at least 48 hours.

4.2. Most important symptoms and effects, both acute and delayed

Headache, nausea, shortness of breath, sore throat, redness on the skin. Repeated or prolonged contact may cause skin sensitization. Repeated or prolonged inhalation exposure may cause asthma.

4.3. Indication of any immediate medical attention and special treatment needed

Depending on the degree of exposure, periodic medical examination is suggested.

5. Firefighting measures

5.1. Extinguishing media

Suitable extinguishing media:

Foam, CO_2 or dry powder. Water spray may be used if no other available and then in copious quantities.

Unsuitable extinguishing media:

High volume water jet.

5.2. Special hazards arising from the substance or mixture

Carbon dioxide, carbon monoxide, hydrogen cyanide, nitrogen oxides, isocyanate vapours. The substances/groups of substances mentioned can be released in case of fire.

5.3. Advice for firefighters

Reaction between water and hot isocyanate may be vigorous (strongly exothermic). Prevent washings from entering watercourses. Keep fire-exposed containers cool by spraying with water.

Special protective equipment:

Fire-fighters should wear appropriate protective equipment and self- contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode. Safety boots, gloves, safety helmet and protective clothing should be worn.

Further information:

In the event of fire and/or explosion do not breathe fumes. Fire in vicinity poses risk of pressure build-up and rupture. Containers at risk from fire should be cooled with water and, if possible, removed from the danger area. Due to reaction with water producing CO_2 gas, a hazardous build-up of pressure could result if contaminated containers are re-sealed. Containers may burst if overheated. Do not allow contaminated extinguishing water to enter the soil, groundwater or surface waters.

6. Accidental release measures

6.1. Personal precautions, protective equipment and emergency procedures

Immediately contact emergency personnel. Evacuate the area. Keep upwind to avoid inhalation of vapours. Clean-up should only be performed by trained personnel. Keep unauthorized persons away.

6.1.1. For non-emergency personnel:

Remove not affected people. Inform the relevant emergency services and authorities.

6.1.2. For emergency responders:

People dealing with major spillages should wear full protective clothing including respiratory protection. Use suitable protective equipment.

6.2. Environmental precautions

Do not allow contaminated extinguishing water to enter the soil, ground-water or surface waters. Avoid dispersal of spilt material and runoff and contact with drains and sewers.

6.3. Methods and material for containment and cleaning up

Absorb spillages onto sand, earth or any suitable adsorbent material. Leave to react for at least 30 minutes. Do not absorb onto sawdust or other combustible materials. Contaminated adsorbent material shall be disposed according to Section 13. Wash the spillage area with water.

6.4. Reference to other sections

Information regarding exposure controls/personal protection and disposal considerations can be found in Section 8 and 13.

7. Handling and storage

7.1. Precautions for safe handling

7.1.1. Protective measures:

Provide sufficient air exchange and/or exhaust in work rooms. In all workplaces of the plant where high concentrations of isocyanate aerosols and/or vapours may be generated (e.g. during pressure release, mould venting or when cleaning mixing heads with an air blast), appropriately located exhaust ventilation must be provided in order to prevent occupational exposure limits from being exceeded. The air should be drawn

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away from the personnel handling the product. The efficiency of the ventilation system must be monitored regularly because of the possibility of blockage. Atmospheric concentrations should be minimised and kept as low as reasonably practicable below the occupational exposure limit.

7.1.2. Advice on general occupational hygiene:

No eating, drinking, smoking or tobacco use at the place of work.

Contact with skin and eyes and inhalation of vapours must be avoided under all circumstances. Keep equipment clean. A basic essential in sampling, handling and storage is the prevention of contact with water. Keep stocks of decontaminant readily available.

7.2. Conditions for safe storage, including any incompatibilities

Store and transport in separate, airtight vessels, between +10 °C and +25 °C. The containers and vessels shall be protected from direct sunshine and other weather impacts. Keep container tightly closed and sealed until ready for use. Containers that have been opened must be carefully resealed and kept upright to prevent leakage. Do not store in unlabelled containers. Use appropriate container to avoid environmental contamination.

7.3. Specific end use(s)

For the relevant identified use(s) listed in Section 1 the advice mentioned in this section is to be observed.

8. Exposure controls/personal protection

8.1. Control parameters

8.1.1. Occupational exposure limits in air

A workplace exposure limit (WEL) of 0.02 mg/m3 for total isocyanates (as NCO) as an 8-hour TWA, and a short term WEL (15 min) of 0.07 mg/m3 have been assigned in the United Kingdom. A BMGV for isocyanates, based on the measurement of urinary diamines, has been set at 1 µmol diamine/mol creatinine.

8.1.2. DNEL/PNEC-values

The risk characterization of PMDI (CAS: 9016-87-9) is the following:

Workers:

Acute/short-term exposure – systemic effects (dermal): DNEL = 50 mg/kg bw/day

Acute/short-term exposure – systemic effects (inhalation): DNEL = 0.1 mg/m^3

Acute/short-term exposure – local effects (dermal): DNEL = 28.7 mg/cm2

Acute/short-term exposure – local effects (inhalation): DNEL = 0.1 mg/m^3

Long-term exposure – systemic effects (inhalation): DNEL = 0.05 mg/m^3

Long-term exposure – systemic effects (dermal): Not applicable.

Long-term exposure – local effects (inhalation): DNEL = 0.05 mg/m^3

Long-term exposure – local effects (dermal): Not applicable.

PNEC sediment: As PMDI is a reactant with water, access of water to PMDI and vice versa is strictly controlled. Furthermore, PMDI polymerizes in the presence of water and thus exposure of PMDI to sediment is highly likely to be negligible. Therefore, PNEC sediment cannot be derived for PMDI.

PNEC soil: 1 mg/kg soil dw

PNEC oral: There are no data on effects of oral PMDI to birds. Exposure to birds is not expected and data from experimental animals show PMDI to be of low oral toxicity.

8.2. Exposure controls

Respiratory protection:

Respiratory protection in case of vapour/aerosol release. Combination filter for organic, inorganic, acid inorganic, and basic gases/vapours (e.g. EN 14387 Type ABEK) shall be used.

Hand protection:

Chemical resistant protective gloves (EN 374)

Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN 374):

butyl rubber (butyl) – 0.7 mm coating thickness

nitrile rubber (NBR) – 0.4 mm coating thickness

chloroprene rubber (CR) – 0.5 mm coating thickness

Unsuitable materials:

polyvinyl chloride (PVC) – 0.7 mm coating thickness

polyethylene (PE) laminate – ca. 0.1 mm coating thickness

Eye protection:

Safety glasses with side-shields (frame goggles) (e.g. EN 166).

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Body protection:

Safety shoes (e.g. according to EN 20346) and closed workwear.

General safety and hygiene measures:

Do not breathe vapour/spray. With products freshly manufactured from isocyanates body protection and chemical resistant protective gloves is recommended. Wearing of closed workwear is required additionally to the personal protective equipment. No eating, drinking, smoking or tobacco use at the place of work. Take off immediately all contaminated clothing. Hands and/or face should be washed before breaks and at the end of the shift. After work the skin should be cleaned and skin-care agents applied.

9. Physical and chemical properties

9.1. Information on basic physical and chemical properties

Appearance	liquid, dark-brown
Odour	damp
Odour threshold	not known
pH-value	not applicable (reacts with water)
Melting point/freezing point	not defined (mixture)
Boiling range	> 200 °C
Flash point	> 200 °C (MDI)
Evaporation rate	not defined (mixture)
Flammability (solid, gaseous)	not applicable (liquid)
Ignitable, explosive range	not defined (mixture)
Vapour pressure	< 0.00001 mbar (at 20 °C)
Vapour density	not defined (mixture)
Density	1.24 ± 0.02 g/cm3 (at 25 °C)
Solubility	reacts with water with slow CO_2 appearance at the border area into non-soluble, high melting point or not melting polyurea
Partition coefficient n-octanol/water	not applicable (mixture)
Self-ignition temperature	4,4'-MDI does not ignite till 601 °C
Decomposition temperature	not applicable (mixture)
Viscosity	190–250 mPas (at 25 °C)
Explosive properties	non-explosive
Oxidising properties	non-oxidizing

9.2. Other information

No data.

10. Stability and reactivity

10.1. Reactivity

Reacts with water, acids, alcohols, amines, bases, and oxidants.

10.2. Chemical stability

The main removal mechanism of MDIs in the environment is hydrolysis. MDI reacts quickly with water to form predominantly solid, insoluble polyureas. Under conditions typical of many types of environmental contact, i.e. with relatively poor dispersion of the isocyanate, the interfacial reaction leads to the formation of a solid crust encasing partially reacted product. This crust restricts ingress of water and egress of amine, and hence slows and modifies hydrolysis.

Stability in organic solvents:

All MDI isomers and forms are highly unstable in dimethyl sulphoxide (DMSO) solvent, water content of the DMSO is increasing breakdown. MDI is more stable in EGDE (ethylene glycol dimethyl ether) solvent.

(Read-across based on 4,4'-methylenediphenyl diisocyanate – CAS 101-68-8.)

10.3. Possibility of hazardous reactions

Reaction is slow with cold or warm water (< 50 °C), with hot water or steam the reaction is faster, producing carbon- dioxide causing pressure increase. Acids, alcohols, amines, bases, and oxidants may cause fire and explosion hazard.

10.4. Conditions to avoid

High temperature, moisture, strong light.

10.5. Incompatible materials

Substances to avoid:

water, acids, alkalis, alcohols, amines.

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10.6. Hazardous decomposition products

No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological information

The mixture has not been tested. Information is related to 4,4'-Methylenediphenyl diisocyanate if no other is mentioned.

11.1. Information on toxicological effects

Acute toxicity – oral:	Harmful
Rats (female)	LD50 = 632 mg/kg Tris (2-chloro-1-methylethyl) phosphate CAS 13674-84-5
Acute toxicity – inhalation (aerosol):	Harmful
Rats	LC50 = 0.49 mg/l air (4 h) OECD Guideline 403
Rats	LC50 > 7 mg/l air (4 h), dusts and mists OECD 403 Acute Inhalation Toxicity / 433 Acute Inhalation Toxicity: Fixed Concentration Procedure Tris(2-chloro-1-methylethyl) phosphate CAS 13674-84-5
Acute toxicity – dermal:	Not classified. Based on available data, the classification criteria are not met.
Rabbit	LD50 > 9400 mg/kg bw (24 h) OECD Guideline 402

11.2. Irritation/Corrosion

Summarized the results of the studies together with human occupational case reports support the official classification.

Skin corrosion/Skin irritation:	Irritating

Irritating in rabbits. (4 h / 14 days)

OECD Guideline 404

Eye damage/Irritation:	—

Not irritating in rabbits. (24 h / 21 days)

OECD Guideline 405

(Read-across based on methylenediphenyl diisocyanate, isomer mixture – CAS 26447-40-5.) Summarized the available animal data would not support classification of MDI as an eye irritant. But together with human occupational case reports in which symptoms of eye irritation were reported the legal classification as eye irritant should be applied.

11.3. Sensitisation

Animal data as well as studies in humans provide evidence of possible skin sensitisation, and of respiratory sensitisation due to MDI. Animal studies indicate that MDI is a strong allergen. Human case reports describe the occurrence of allergic contact dermatitis due to MDI exposure.

Skin sensitisation:	Mice Sensitizing OECD Guideline 429 (LLNA)
Respiratory sensitisation:	Rats (male) Sensitizing OECD Guideline 39

11.4. Germ cell mutagenicity

Not classified. Based on available data, the classification criteria are not met.

11.5. Carcinogenicity

Carc. 2

Rats (inhalation, aerosol):	NOAEC = 0.2 mg/ m³ air (toxicity) (2 years; 6 h/day, 5 days/week) NOAEC = 1 mg/m³ air (carcinogenicity) (2 years; 6 h/day, 5 days/week) LOAEC = 6 mg/m³ air (carcinogenicity) (2 years; 6 h/day, 5 days/week) OECD Guideline 453

11.6. Reproductive toxicity

Not classified. Based on available data, the classification criteria are not met.

Effects on fertility:	No fertility, nor multigeneration studies are available.
Rats (inhalation):	NOAEL = 4 mg/m³ air (developmental toxicity) (10 days; 1/day, 6 h) NOAEL = 4 mg/m³ air (maternal toxicity) (10 days; 1/day, 6 h) OECD Guideline 414

11.7. STOT – single exposure

MDI is irritant to the respiratory tract.

11.8. STOT – repeated exposure

Harmful

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Rats (inhalation, aerosol):	NOAEC = 0.2 mg/m3 air (2 years; 6 h/day, 5 days/week) LOAEC = 1.0 mg/m3 air (2 years; 6 h/day, 5 days/week) Target organs: respiratory – lung OECD Guideline 453

11.9. Aspiration hazard

Not classified due to lack of data.

12. Ecological information

The mixture has not been tested. Information is related to 4,4'-Methylenediphenyl diisocyanate if no other is mentioned.

12.1. Toxicity

12.1.1. Aquatic toxicity

Short-term toxicity to fish

Freshwater fish (Danio rerio): LC50 > 1000 mg/l (96 h)

OECD Guideline 203

Danio rerio (zebrafish): LC50 = 56,2 mg/l (96 h)

Tris(2-chloro-1-methylethyl) phosphate, CAS: 13674-84-5

Long-term toxicity to fish

Data waiving. In accordance with column 2 of REACH Annex IX the long-term toxicity testing on fish shall be proposed if the chemical safety assessment according to Annex I indicates the need to investigate further the effects on aquatic organisms. The corresponding PEC/PNEC ratios would be less than 1. Taking into account the scientific and exposure arguments, it appears appropriate to waiver the long-term fish/plant/soil and sediment toxicity studies.

Short-term toxicity to aquatic invertebrates

Freshwater invertebrates (Daphnia magna): EC50 > 1000 mg/l (24 h)

OECD Guideline 202

Daphnia magna: EC50 = 131 mg/l (48 h)

Tris(2-chloro-1-methylethyl) phosphate, CAS: 13674-84-5

Long-term toxicity to aquatic invertebrates

Freshwater invertebrates (Daphnia magna): NOEC ≥ 10 mg/l (21 days)

OECD Guideline 211

Toxicity to aquatic algae and cyanobacteria

Freshwater algae (Desmodesmus subspicatus) EC50 > 1640 mg/l (72 h)

OECD Guideline 201

Freshwater algae (Desmodesmus subspicatus) EC50 = 82 mg/l (72 h)

Tris(2-chloro-1-methylethyl) phosphate, CAS: 13674-84-5

Toxicity to aquatic plants other than algae

Data waiving. Not required by REACH annexes. However, a mesocosm study with PMDI exists in which the toxicity towards macrophytes (Potamogeton crispus and Zannichellia palustris) was assessed. No toxicity was observed at a loading of 1000 and 10000 mg/l, approximately 100% of the substance was found in the sediment as hardened material.

Toxicity to microorganisms

Microorganisms (activated sludge) EC50 > 100 mg/l (3 h)

OECD Guideline 209

Toxicity to other aquatic organisms

This information is not available, but not required under REACH.

12.1.2. Sediment toxicity

Data waiving. Annex X states that this study need not be conducted if the chemical safety assessment does not indicate a need to further investigate the effects on sediment organisms.

12.1.3. Terrestrial toxicity

Toxicity to soil macroorganisms except arthropods

Eisenia fetida LC50 > 1000 mg/kg soil dw (14 days)

OECD Guideline 207

Toxicity to terrestrial arthropods

Data waiving. Based on the chemical safety assessment and the risk assessment, there is no need to further investigate the terrestrial arthropods toxicity as there is no risk for the terrestrial environment as indicated by the PEC/PNEC ratio being < 0.239. Direct/indirect exposure to soil is unlikely.

Toxicity to terrestrial plants

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Avena sativa EC50 > 1000 mg/kg soil dw (14 days)

Lactuca sativa EC50 > 1000 mg/kg soil dw (14 days)

OECD Guideline 208

Toxicity to soil microorganisms

Data waiving. Annex X states that this study need not be conducted if the chemical safety assessment does not indicate a need to further investigate the effects on sediment organisms.

Toxicity to other above-ground organisms

Data waiving. Not required by REACH annexes.

12.1.4. Conclusion on classification

Hazardous to the aquatic environment (acute)

Based on available data, the classification criteria are not met.

(EC/LC50 for fish, invertebrates and algae > 1000 mg/l)

Hazardous to the aquatic environment (chronic)

Based on available data, the classification criteria are not met.

(NOEC for algae > 1640 mg/l; NOEC for invertebrates > 10 mg/l)

12.2. Persistence and degradability

Phototransformation in air

Half-life (DT50): 0.92 days

Hydrolysis

MDI reacts with water to form predominantly inert polyurea.

Half-life (DT50): ca. 20 h (at 25 °C)

(Read-across based on oligomeric MDI – CAS 32055-14-4)

Phototransformation in water and soil

No data is available.

Biodegradation in water

Under test conditions no biodegradation observed. (28 days)

OECD Guideline 302C

Biodegradation in water and sediment

Data waiving. In accordance with Annex XI, simulation biodegradation tests are technically not feasible as the test substance reacts quickly with water. The corresponding PEC/PNEC ratios would be less than 1. Taking into account the scientific and exposure arguments, it appears appropriate to waiver the long-term fish/plant/soil and sediment toxicity studies.

Biodegradation in soil

Data waiving. See at Biodegradation in water and sediment.

12.3. Bioaccumulative potential

Bioaccumulation – aquatic/sediment

Due to the high reactivity of the substances of the MDI category with water, bioaccumulation tests can in principle not be performed with these substances. However, one bioaccumulation test with 4,4'-MDI and a mesocosm study with PMDI with an indication of bioaccumulation potential have been performed. As no analytical measurements were done, it cannot be determined if the values are truly related to MDI. However, based on the available information and the reactivity of MDI substances of the category approach, no new bioaccumulation study is deemed necessary.

BCF (Cyprinus carpio) 200 (28 days)

Method: OECD Guideline 305E

Terrestrial bioaccumulation

No data is available on terrestrial bioaccumulation, but it is not required under REACH.

12.4. Mobility in soil

Adsorption/desorption

Data waiving. According to Annex VIII the study need not be done if the test substance degrades rapidly. The corresponding PEC/PNEC ratios would be less than 1. Taking into account the scientific and exposure arguments, it appears appropriate to waiver the long-term fish/plant/soil and sediment toxicity studies.

Volatilisation

The Henry's Law Constant, estimated from the measured vapour pressure and the calculated water solubility, is 2.263 x 10-7 atm-m3/mole. Hence, volatilization is unlikely to be a significant removal mechanism for MDI substances of the category approach.

12.5. Results of PBT and vPvB assessment

Conclusion for the P criterion

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The results from the biodegradation test indicate that PMDI is not biodegradable. Based on experimental hydrolysis and indirect photolysis half-lives, PMDI is not considered to be persistent in the environment and is identified as not P. Based on the justification in the category approach, it is assumed that all MDI analogues included in the category are not P.

Conclusion for the B criterion

Although MDI has a high measured log Pow value (4.51), a full bioaccumulation test with 4,4'-MDI indicated that the bioaccumulation potential is low. Due to the fast hydrolysis, exposure of the environment to the substance is unlikely or very low, there is no potential for significant bioaccumulation possible. Hence, 4,4'-MDI does not fulfil the requirements for the B criterion and is not identified as B. Based on the justification in the category approach, it is assumed that all MDI analogues included in the category are not B.

Conclusion for the T criterion

The concentrations tested were far above the water solubility of the MDI substances (7.5 mg/l). However, the water solubility limit of MDI is far above the criteria for T and on the basis of aquatic toxicity tests MDI is identified as not T. However, according to Annex I of 67/548/EEC MDI is classified as Xn, R48, which automatically triggers a T. Based on this classification MDI is identified as T.

12.6. Other adverse effects

It is not expected that substance has an effect on global warming, ozone depletion in the stratosphere or ozone formation in the troposphere.

Secondary poisoning

Based on the available information, there is no indication of a bioaccumulation potential and, hence, secondary poisoning is not considered relevant. Exposure to birds is not expected and data from experimental animals show MDI to be of low oral toxicity.

13. Disposal considerations

13.1. Waste treatment methods

The products becoming useless and the contaminated containers not suitable for product storage must be handled as hazardous waste in accordance with EU and regional hazardous waste regulations. European Waste Catalogue code: 08 05 01

13.1.1. Product / Packaging disposal:

Contaminated packaging should be emptied as far as possible; than it can be passed on for recycling after being thoroughly cleaned. Wrappings cleaned from contamination with suitable cleaning process (e.g. by steaming, treating with washing fluid, etc.) must be considered as non-hazardous waste.

13.1.2. Waste treatment options:

Incinerate in suitable incineration plant, observing local authority regulations.

14. Transport information

Land transport (ADR/RID/GGVSE)

Sea transport (IMDG Code/GGVSee)

Air transport (ICAO-IATA/DGR)

14.1. UN number

Not dangerous goods

14.2. UN proper shipping name

Not dangerous goods

14.3. Transport hazard class(es)

Not dangerous goods

14.4. Packing group

Not dangerous goods

14.5. Environmental hazards

Marine pollutant: no

14.6. Special precautions for users

EmS number: Not dangerous goods

14.7. Transport in bulk according to Annex II of Marpol and the IBC Code

Not relevant

15. Regulatory information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

15.1.1. Information regarding relevant EU safety, health and environmental provisions

ISOPA, the European Diisocyanate & Polyol Producers Association has elaborated a Guideline document for the safe treatment of MDI containing products. The Guidelines have been built into this data sheet.

15.2. Chemical safety assessment

In accordance with REACH chemical safety assessment (CSA) has not been carried out for the product. However, the results from the CSA for 4,4'-MDI were transposed into this SDS.

16. Other information

The information given corresponds with our actual knowledge and experience. This information is meant to describe our product in view of possible safety requirements. Classification of the mixture is based on the classification of components.

16.1. Indication of changes

This is the first edition of the SDS.

16.2. Abbreviations and acronyms

BCF	Bioconcentration factor
BMGV	Biological monitoring guidance value
bw	bodyweight
CAS No.	Chemical Abstracts Service number
CLP	Regulation on classification, labelling and packaging
DNEL	Derived no effect level
dw	dry weight
EC No.	EINECS and ELINCS number
EC50	Half maximal effective concentration
EEC	European Economic Community
EINECS	European Inventory of Existing Commercial Chemical Substances
ELINCS	European List of Notified Chemical Substances
EU	European Union
LC50	Lethal concentration, 50 %
LD50	Median lethal dose
LLNA	Local lymph node assay
LOAEC	Lowest Observed Adverse Effect Concentration
NOAEC	No Observed Adverse Effect Concentration
NOAEL	No Observed Adverse Effect Level
NOEC	No Observed Effect Concentration
OECD	Organisation for Economic Cooperation and Development
PBT	Persistent, Bioaccumulative and Toxic
PEC	Predicted Environmental Concentration
PMDI	Polymeric MDI (CAS: 9016-87-9)
PNEC	Predicted No Effect Concentration
REACH	The Registration, Evaluation, Authorisation and Restriction of Chemicals
SDS	Safety Data Sheet
TWA	Time-weighted average
vPvB	Very Persistent and Very Bioaccumulative
WEL	Workplace exposure limit

16.3. Key literature references and sources for data

Safety data sheets, received from the raw materials suppliers.

16.4. Full text of abbreviations

H-Phrases

H302	Harmful if swallowed.
H315	Causes skin irritation.
H317	May cause an allergic skin reaction.
H319	Causes serious eye irritation.
H332	Harmful if inhaled.
H334	May cause allergy or asthma symptoms or breathing difficulties if inhaled.
H335	May cause respiratory irritation.
H351	Suspected of causing cancer.
H373	May cause damage to organs through prolonged or repeated exposure: respiratory system, inhalation.

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P-Phrases

P260	Do not breathe dust/fume/gas/mist/vapours/spray.
P280	Wear protective gloves/protective clothing/eye protection/face protection.
P284	Wear respiratory protection.
P302+P352	IF ON SKIN: Wash with plenty of soap and water.
P304+P340	IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P305+P351+P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses if present and easy to do. Continue rinsing.
P308+P313	If exposed: Call a POISON CENTER or doctor/physician.

Hazard classes

Acute Tox.	Acute toxicity
Carc.	Carcinogenicity
Eye Irrit.	Serious eye irritation
Resp. Sens.	Respiratory sensitization
Skin Irrit.	Skin irritation
Skin Sens.	Skin sensitization
STOT RE	Specific target organ toxicity – repeated exposure
STOT SE	Specific target organ toxicity – single exposure

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Safety data sheet

LR Silicate Resin Type Summer

"A" component for water glass – polyisocyanate based two-component synthetic resin. The synthetic resin (components "A" + "B") is used for the lining

HAZARD OVERVIEW · GHS / CLP

Danger

H315 Causes skin irritation. H318 Causes serious eye damage.
P262 Do not get in eyes, on skin, or on clothing. P280 Wear protective gloves/protective clothing/eye protection/face protection. P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if P303+P361+P353 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with

PRODUCT IDENTIFICATION

PRODUCT
LR Silicate Resin Type Summer

IDENTIFIED USE
"A" component for water glass – polyisocyanate based two-component synthetic resin. The synthetic resin (components "A" + "B") is used for the lining of sewer pipes and manholes. The application has to be carried out under professional, industrial conditions by persons having proper previous training.

MANUFACTURER & EMERGENCY

MANUFACTURER
UAB Lateral Repairs
Paberžių g. 5, Tauragė, LT-72328, Lithuania

CONTACT
info@lateralrepairs.com · +370 698 76581

EMERGENCY
Regional Medicines and Poisons Information Centre NI, Belfast Tel.: +44 844 892 0111 (24 hrs)

DOCUMENT	REGULATION	VERSION	DATE OF ISSUE
Safety Data Sheet	1907/2006 · 2015/830	1.0 / EN	01/06/2020

1. Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

LR Silicate Resin Type Summer

1.2. Relevant identified uses of the substance or mixture and uses advised against

"A" component for water glass – polyisocyanate based two-component synthetic resin. The synthetic resin (components "A" + "B") is used for the lining of sewer pipes and manholes. The application has to be carried out under professional, industrial conditions by persons having proper previous training.

1.3. Details of the supplier of the safety data sheet

Producer/Supplier:

UAB Lateral Repairs

Street/POB:

Paberziu g. 5

Postcode/City/Country:

LT-72328, Taurage, Lithuania

E-mail address for a competent person responsible for the safety data sheet:

info@lateralrepairs.com

Phone:

+370 698 76581

1.4. Emergency telephone number

Regional Medicines and Poisons Information Centre NI, Belfast

Tel.: +44 844 892 0111 (24 hrs)

2. Hazards identification

2.1. Classification of the substance or mixture

Classification according to Regulation (EC) No 1272/2008 (CLP):

Hazard class / category	Code	Hazard statement
Skin Irrit. 2	H315	Causes skin irritation.
Eye Dam. 1	H318	Causes serious eye damage.

2.2. Label elements

Labelling according to Regulation (EC) No 1272/2008 (CLP)

Hazard pictograms:

Signal word:

Danger

Hazard statements:

H315 Causes skin irritation.

H318 Causes serious eye damage.

Precautionary statements:

P262 Do not get in eyes, on skin, or on clothing.

P280 Wear protective gloves/protective clothing/eye protection/face protection.

P303+P361+P353 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower.

P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

Hazard determining component(s) for labelling:

Silicic acid, sodium salt

2.3. Other hazards

The substances in the mixture do not meet the PBT/vPvB criteria according to REACH, Annex XIII.

3. Composition/information on ingredients

3.1. Mixtures

Chemical characterization

Substance	EC No.	CAS No.	REACH Reg. No.	Content (%)	Classification (CLP)[1]
Silicic acid, sodium salt (Molar ratio $Na_2O : SiO_2$ = 1 : > 1.6 – < 2.6)	215-687-4	1344-09-8	01-2119448725-31	25–50	Skin Irrit. 2 (H315), Eye Dam. 1 (H318)
Water	231-791-2	7732-18-5	—	50–75	—

[1] – See Section 16 for the full text of the abbreviations declared above.

4. First aid measures

4.1. Description of first aid measures

General information:

No special measures necessary. When in doubt or if symptoms are observed, get medical advice. First aid responders should pay attention to self-protection.

4.1.1. Inhalation:

Remove to fresh air. Obtain medical attention if breathing difficulty persists.

4.1.2. Skin contact:

Soiled, fairly soaked clothing must be immediately removed. In case of contact with skin, wash off immediately with plenty of water. Do not allow the product to dry on the skin. Consult a doctor if skin irritation persists.

4.1.3. Eye contact:

Immediately wash affected eyes carefully, but thoroughly for at least 15 minutes under running water with eyelids held open. Use an eyewash station if possible. Consult an eye specialist.

4.1.4. Ingestion:

Immediately rinse mouth and drink plenty of water. Do not induce vomiting. Never give anything by mouth to an unconscious person or a person with cramps. Call a physician immediately.

4.2. Most important symptoms and effects, both acute and delayed

No symptoms known so far.

4.3. Indication of any immediate medical attention and special treatment needed

Hints for the physician/dangers:

The product contains alkali silicate. After swallowing and vomiting aspiration into lung is possible, which can cause chemical pneumonia or asphyxia.

5. Firefighting measures

5.1. Extinguishing media

Suitable extinguishing media:

The product itself is non-combustible. Adapt fire extinguishing measures to surrounding areas. Water spray, carbon dioxide, dry chemical extinguishing agent (powder), foam.

Non-suitable extinguishing media:

full water jet

5.2. Special hazards arising from the substance or mixture

None known.

5.3. Advice for firefighters

Special protective equipment:

Use suitable breathing apparatus. In case of fire and/or explosion do not breathe fumes.

Further information:

Fire in vicinity poses risk of pressure build-up and burst. Containers at risk from fire should be cooled with water spray. Fire residues and contaminated extinguishing water should be disposed of in accordance with local authority regulations. Carbon dioxide should be applied carefully in confined spaces since it can displace oxygen.

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6. Accidental release measures

6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective clothing. Avoid contact with skin, eyes, and clothing. High risk of slipping due to leakage/spillage of product.

6.2. Environmental precautions

Do not allow to enter drains or waterways. Prevent spread over a wide area (e.g. by containment or oil barriers).

6.3. Methods and material for containment and cleaning up

Take up with absorbent material (e.g. sand, kieselguhr, universal binder). Rinse away rest with plenty of water. Dispose according to authority regulations.

6.4. Reference to other sections

Safe handling:

see Section 7

Personal protective equipment:

see Section 8

Disposal:

see Section 13

7. Handling and storage

7.1. Precautions for safe handling

Observe the usual precautions for handling chemicals. Open and handle container with care. Avoid contact with skin, eyes and clothing. Do not breathe aerosol. Wear personal protective equipment. The product is not combustible or explosive, but heating leads to pressure build-up and risk of burst.

7.2. Conditions for safe storage, including any incompatibilities

Requirements for storage rooms and vessels: Keep only in the original container. Keep container tightly closed. Provide adequate protection against leakage, e.g. with the aid of collection trays or lowered areas. Provide a solvent-resistant and solid floor.

Further information on storage conditions:

Protect from frost.

Recommended storage temperature:

between +5 and +45 °C

Storing together hints:

Keep away from food, drink, and feeding stuff. Do not store with acids.

Storage stability:

Under correct storing conditions the product is stable for at least 12 months.

7.3. Specific end use(s)

Only for industrial users.

8. Exposure controls/personal protection

8.1. Control parameters

Limit values according to REACH registration:

DNEL for workers: 5.61 mg/m3 (inhalation); 1.59 mg/kg bw/day (dermal)

PNEC for freshwater: 7.5 mg/L, for sewage treatment plant: 348 mg/L

8.2. Exposure controls

General protective and hygiene measures:

Observe the usual precautions when handling chemicals. Avoid contact with skin and eyes. Do not breathe aerosol. Do not eat, drink or smoke during working time. Take off immediately all contaminated clothing and wash before reuse. Hands and/or face should be washed before breaks and at the end of the shift. Take a shower, if necessary. Apply skin care products after work.

Occupational exposure controls:

Respiratory protection:

Usually no personal respiratory protection is needed. It is only required when the exposure limit value is exceeded, or in the event of aerosol/mist formation. Suitable respiratory protection apparatus: P2 particle filter device (DIN EN 143).

Hand protection:

Tested protective gloves must be worn (DIN EN 374). Suitable material: NBR (nitrile rubber), butyl rubber. Thickness >= 0.4 mm, breakthrough time (maximum wearing time) >= 480 min. Observe the wear time limits as specified by the manufacturer.

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Eye protection:

Safety glasses with side protection (DIN EN 166).

Body protection:

Clothing as usual in the chemical industry.

Environmental exposure controls:

See Section 7. No additional measures necessary.

9. Physical and chemical properties

9.1. Information on basic physical and chemical properties

Appearance	liquid, dark brown
Odour	odourless
Odour threshold	not applicable
pH-value	13-14
Melting point/freezing point	no data
Boiling range	> 100 °C
Flash point	does not ignite
Evaporation rate	no data
Flammability (solid, gaseous)	not applicable
Ignitable, explosive range	not applicable
Vapour pressure	no data
Vapour density	no data
Density	1.5 ± 0.1 g/cm3 (25 °C)
Solubility	completely miscible with water
Partition coefficient n-octanol/water	not applicable
Self-ignition temperature	not applicable
Decomposition temperature	not applicable
Dynamic viscosity	400 ± 50 mPa.s (25 °C)
Explosive properties	non-explosive
Oxidizing properties	non-oxidizing

9.2. Other information

Solids content: ca. 46%

10. Stability and reactivity

10.1. Reactivity

Under normal conditions of storage and use, hazardous reactions will not occur.

10.2. Chemical stability

Under normal conditions of storage and use, hazardous reactions will not occur.

10.3. Possibility of hazardous reactions

Under normal conditions of storage and use, hazardous reactions will not occur.

10.4. Conditions to avoid

Protect from frost.

10.5. Incompatible materials

Substances to avoid:

acids. Reacts with metals like aluminium, tin, zinc under evolution of hydrogen and heat.

10.6. Hazardous decomposition products

No hazardous decomposition products if stored and handled as prescribed/indicated (see Section 7).

11. Toxicological information

11.1. Information on toxicological effects

Acute toxicity

Based on available data, the classification criteria are not met.

Silicic acid, sodium salt (Molar ratio $Na_2O : SiO_2$ = 1 : > 1.6 – < 2.6) · CAS 1344-09-8

Exposure route	Dose	Species	Source
oral	LD50 > 2000 mg/kg	Rat	IUCLID
dermal	LD50 > 5000 mg/kg	Rat	IUCLID

Irritation and corrosivity	Causes skin irritation. Causes serious eye damage.
Sensitizing effects	Based on available data, the classification criteria are not met.
Carcinogenic/mutagenic/toxic effects for reproduction	Based on available data, the classification criteria are not met.
STOT – single exposure	Based on available data, the classification criteria are not met.
STOT – repeated exposure	Based on available data, the classification criteria are not met.
Aspiration hazard	Based on available data, the classification criteria are not met.

12. Ecological information

The data refer to the product with the given composition and were used cross-referenced.

12.1. Toxicity

The product has not been tested.

Silicic acid, sodium salt (Molar ratio $Na_2O : SiO_2$ = 1 : > 1.6 – < 2.6) · CAS 1344-09-8

Aquatic toxicity	Dose	Time	Species	Source
Acute fish toxicity	LC50 1108 mg/l	96 h	Brachydanio rerio (zebrafish)	IUCLID
Acute algae toxicity	ErC50 207 mg/l	72 h	Scenedesmus subspicatus	IUCLID
Acute crustacea toxicity	EC50 1700 mg/l	48 h	Daphnia magna (big water flea)	IUCLID

12.2. Persistence and degradability

No data available.

12.3. Bioaccumulative potential

No data available.

12.4. Mobility in soil

No data available.

12.5. Results of PBT and vPvB assessment

The substances in the mixture do not meet the PBT/vPvB criteria according to REACH, Annex XIII.

12.6. Other adverse effects

The product is an alkali. Before discharge into sewage plants the product normally needs to be neutralised.

Further information:

Do not allow uncontrolled discharge of product into the environment. Do not allow to enter soil, waterways or waste water canal.

13. Disposal considerations

13.1. Waste treatment method

Advice on disposal:

Dispose according to legislation. Consult the appropriate local waste disposal expert about waste disposal.

Waste disposal number of waste from residues/unused products:

060205 WASTES FROM INORGANIC CHEMICAL PROCESSES; wastes from the MFSU of bases; other bases; hazardous waste

Contaminated packaging:

Handle contaminated packages in the same way as the substance itself. Packaging which cannot be properly cleaned must be disposed of. Non- contaminated packages may be recycled. Do not mix with other wastes.

14. Transport information

Land transport (ADR/RID) / Inland waterways transport (ADN)/ Marine transport (IMDG) / Air transport (ICAO-TI/IATA-DGR)

14.1. UN number

No dangerous good in sense of this transport regulation.

14.2. UN proper shipping name

No dangerous good in sense of this transport regulation.

14.3. Transport hazard class(es)

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No dangerous good in sense of this transport regulation.

14.4. Packing group

No dangerous good in sense of this transport regulation.

14.5. Environmental hazards

ENVIRONMENTALLY HAZARDOUS: no Danger releasing substance: No dangerous good in sense of this transport regulation.

14.6. Special precautions for user

No dangerous good in sense of this transport regulation.

14.7. Transport in bulk according to Annex II of Marpol and the IBC Code

No dangerous good in sense of this transport regulation.

15. Regulatory information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

EU regulatory information:

2010/75/EU (VOC): 0%

Information according to 2012/18/EU (SEVESO III): Not subject to 2012/18/EU (SEVESO III)

Additional information:

The product does not contain substances of very high concern (SVHC).

National regulatory information:

Water contaminating class (D): 1 – slightly water contaminating

15.2. Chemical safety assessment

In accordance with REACH chemical safety assessment has not been carried out for the product.

16. Other information

The above information describes exclusively the safety requirements of the product and is based on our present- day knowledge. The information is intended to give you advice about the safe handling of the product named in this safety data sheet, for storage, processing, transport and disposal. The information cannot be transferred to other products. In the case of mixing the product with other products or in the case of processing, the information on this safety data sheet is not necessarily valid for the new made-up material. The information is based on present level of our knowledge. It does not, however, give assurances of product properties and establishes no contract legal rights.

16.1. Indication of changes

The complete data sheet is considered new.

16.2. Abbreviations and acronyms

bw	bodyweight
CAS No.	Chemical Abstracts Service number
CLP	Regulation on Classification, Labelling and Packaging
DNEL	Derived no effect level
EC	European Commission
EC No.	EINECS and ELINCS number
EC50	Half maximal effective concentration
ErC50	EC50 based on growth rate
EINECS	European Inventory of Existing Commercial Chemical Substances
ELINCS	European List of Notified Chemical Substances
EU	European Union
LC50	Lethal concentration, 50%
LD50	Median lethal dose
MFSU	Manufacture, formulation, supply, and use
PBT	Persistent, Bioaccumulative and Toxic
PNEC	Predicted No Effect Concentration
REACH	The Registration, Evaluation, Authorisation and Restriction of Chemicals (EU regulation)
SVHC	Substances of Very High Concern
VOC	Volatile Organic Compound
vPvB	Very Persistent and Very Bioaccumulative

H-Phrases

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H315	Causes skin irritation.
H318	Causes serious eye damage.

P-Phrases

P262	Do not get in eyes, on skin, or on clothing.
P280	Wear protective gloves/protective clothing/eye protection/face protection.
P303+P361+P353	IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower.
P305+P351+P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

Hazard classes

Eye Dam.	Serious eye damage
Skin Irrit.	Skin irritation

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